41004 AI/Analytics Capstone Project

SCHOOL IMMUNIZATION TRACKING SYSTEM

Part 3

Brunda Varshini Katapalli (with 5 team members)

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1. Executive Summary

Note: This report describes a student capstone project. All data in it is synthetic, generated for demonstration, and no real health records were used. Public school names appear only as labels and the figures shown do not describe those schools. The client is described generically as a state health department.

School immunization programs rely on accurate, timely, and accessible data so that vaccination coverage can be monitored across schools, vaccine types, year groups, and Local Health Districts. The state health department currently uses Excel-based files to manage this reporting process. While this approach can support basic reporting, it does not provide a consistent, centralized, or automated way to collect, store, and analyse school immunization data on a scale. As a result, important patterns such as low coverage, incomplete visits, and vaccine-specific gaps may not be identified quickly enough to support timely operational decision-making.

To address this problem, our group developed a prototype School Immunization Tracking System that combines standardized digital submission, structured backend storage, and interactive dashboard reporting. The final prototype uses Microsoft Forms for data entry, Excel on OneDrive for response capture, Power Automate for row detection and transfer, Azure SQL for staging and relational storage, and Power BI for visual reporting. Together, these components demonstrate how the state health department could move from a fragmented spreadsheet-based process toward a more structured and proactive reporting workflow.

The findings of the project confirm that this business problem is significant. The report shows that school-delivered vaccination coverage has declined in recent years, and the prototype data further highlights a large gap between current coverage and herd immunity thresholds. In particular, the synthetic benchmark data used in the project places HPV coverage at approximately 52%, which is well below the expected threshold. The dashboard and model were designed to make these issues easier to identify through threshold-based monitoring, school-level comparison, multi-year analysis, and operational follow-up views.

A major outcome of the project was the successful implementation of the prototype backend workflow. The group developed a staging-table-based pipeline, field mapping between the response file and the database, validation checks, row-level security logic, and Power BI reporting pages aligned to the operational needs of the state health department. The prototype confirmed that submitted data can be captured, transferred, structured, and prepared for reporting in a more reliable and scalable way than the client's current manual approach.

Based on these findings, we recommend that the state health department continue toward a structured digital reporting model. The strongest next steps are to replace the prototype inputs with live operational data, strengthen validation and governance controls, finalize service-side deployment, and continue refining the dashboard to support ongoing immunization monitoring. Overall, this project shows that the state health department can shift from reactive spreadsheet-based reporting toward a more proactive and data-driven school immunization tracking system.

2. The Business Problem

2.1 What is the Business Problem?

School immunization programs rely on accurate and timely data so that vaccination coverage can be monitored across schools, vaccine types, year groups, and Local health Districts. Currently, the state health department uses Excel-based files to track school immunization data across New South Wales. Though Excel can be useful for basic reporting, it can become increasingly difficult to manage as the volume and complexity of the immunization data grows.

The main business problem is that the current spreadsheet-based process does not provide a consistent, centralized, or automated way to collect, store, and analyze school immunization data. Furthermore, this results in several limitations with the current system:

1. **Inconsistent data collection processes –** due to the lack of a standardized digital submission method.
2. **Disconnected and unstructured data storage –** because the data is spread across separate spreadsheets rather than being stored in one neatly structured database.
3. **Limited visibility into immunization performance –** requiring staff to manually review and compare data before identifying trends or problem areas.
4. **Delayed and error-prone trend identification –** making it harder to quickly detect low coverage rates, incomplete visits, or schools that could require follow-up action.

Moreover, the state health department may not be able to identify important patterns quickly enough to support timely decision-making with the current system in place. These patterns include declining vaccination coverage, differences between schools or districts, vaccine-specific gaps, low consent rates, and missed school visits. To address this, the project focused on improving how school immunization data could be collected, structured, and visualized, to ensure that key insights are accessible in an efficient manner.

2.2 Why Does it Matter to the Client?

Our client is a state health department, represented by the project supervisor. This problem holds relevance to the client given that school immunization programs play an important role in protecting students as well as the wider community from vaccine-preventable diseases. The NSW school immunization program currently includes key vaccines such as HPV and dTpa for Year-7 students, and Men ACWY for Year-10 students. In order for these programs to be effective, the state health department needs clear visibility regarding which schools are currently meeting coverage expectations, and which schools may require additional support.

The current spreadsheet-based process limits this visibility and creates several issues for our client, being:

1. **Public health impact -** The current declining vaccination coverage rates pose a risk to not only the health of the students, but the community as well. When a school falls below target thresholds, it is vital that this is identified in a timely fashion in order to allow for the appropriate course of action and resource allocation.
2. **Operational inefficiency -** The current state health department staff are required to spend time manually reviewing the excel spreadsheets which delays their ability to act on the insights gathered thus reducing the effectiveness and efficiency of the immunization program.
3. **Scalability -** Given that the data spans multiple schools, 8 local health districts, unique year groups, and different vaccine types, the volume and complexity of the information involved makes the current manual approach increasingly harder to sustain as the data volume continues to grow.
4. **Reporting Limitations -** One of the requirements of the state health department immunization program is the ability to compare vaccination coverage rates across schools, districts, and years, in order to make informed decisions regarding resource allocation and scheduling. With the current system setup, this is a task which is not readily supported due to the lack of structured reporting and visualisation capabilities within the current Excel-based system
5. **Delayed decision-making -** Given that between the years 2022 to 2024, a declining vaccination trend has already been identified, it is evident that the necessity of a timely and accurate monitoring system is an immediate concern rather than a hypothetical scenario.

For the state health department, the issue is therefore not only about improving data storage, but also about improving the speed, accuracy, and usefulness of the immunization reporting to ensure that staff can identify risks earlier, prioritize follow-up action, and make more informed operational decisions.

2.3 Group's Understanding of the Problem

Upon assessing the current Excel-based system issues, our team determined the main problems to be the data pipeline and visualisation limitations. Although the immunization data already exists within the state health department, the current system lacks efficiency and automation when it comes to collecting, storing, and analysing the data in a way which supports informed decision making. Due to the systems lack of capabilities, this results in potentially critical patterns – such as the declining vaccination rates in specific schools or districts – potentially being missed or not detected in a timely manner.

To address this problem, our team has framed this as a data mining problem, where our goal was to design and demonstrate a system capable of extracting meaningful and actionable insights from immunization data. The objective is to support the operational needs of the state health department in a timely fashion without requiring much manual intervention. This was approached by applying a range of data mining techniques to the dataset, consisting of the following:

1. **Classification –** Categorize schools based on vaccination coverage levels. This enables health teams to quickly identify schools that fall below the target thresholds and may require additional visits.
2. **Clustering –** Group schools with similar immunization trends together. This allows for the identification of common challenges such as low consent rates across regions or demographics.
3. **Association –** Explore the potential relationships between variables within the dataset. This enables the ability to determine whether factors such as the number of scheduled immunization visits influences the coverage rates or whether certain year groups or vaccine types can be linked to lower participation.
4. **Prediction –** Utilize historical immunization data in order to forecast potential future vaccination trends. This allows the state health department to proactively anticipate immunization coverage trends and plan resource allocation more effectively ahead of time.

As the project continued to develop, our understanding of these techniques became more practical. Rather than simply treating the project as a theoretical data mining task, we also began to focus on and consider how these ideas could potentially support a working decision-support system. As an example, classification can be reflected through the use of coverage status indicators, association can be explored through the relationship between consent rates and vaccination coverage, and multi-year analysis can help with identifying trends which should be monitored over time. These developed understandings allow the project to align closer with the state health department's operational needs given that the goal is not only to analyse data but also make important patterns more recognizable and easier to address.

Overall, our group determined the main drawbacks of the state health department's current immunization tracking system as issues related to accuracy, time efficiency, and usability. Our developed solution aims to directly target all three of these pain points through the implementation of standardized digital submission forms, a structured relational database, and a live interactive dashboard. Collectively, these components can support the state health department by shifting the immunization program from a manual reporting process which seems to be reactive, into a proactive data-driven system for school immunization monitoring.

2.4 Herd Immunity Thresholds

The following table outlines the herd immunity thresholds for each vaccine tracked in the NSW school immunization program. These thresholds represent the **minimum population coverage required to prevent widespread disease transmission.**

Vaccine	Disease	Herd Immunity Threshold
HPV	HPV-related cancers	~ 80 - 85% coverage needed
dTpa	Diphtheria, Tetanus, Pertussis	~ 92 - 95% for pertussis
Men ACWY	Meningococcal disease	~ 85 - 90%

Source: World Health Organization (WHO) immunization guidelines & Australian Immunization Handbook (2024).

When coverage falls below these thresholds, unvaccinated individuals, including those who cannot be vaccinated for medical reasons, may be left unprotected. This makes accurate and timely tracking of school-level coverage not just an operational priority, but a public health necessity.

In our final project, these thresholds help guide how vaccination coverage was interpreted within the dashboard. Rather than simply treating all coverage percentages as equal, the system utilizes threshold-based thinking in order to identify schools and vaccine areas which may possibly require attention. This was important because a school with low coverage is not only a reporting issue, but also a potential public health risk which may require follow-up visits, improved consent collection, or even targeted communication with families in surrounding communities.

2.5 Why This Project Addresses a Real Public Health Gap?

Coverage data on its own is an insufficient tool, as it does not reveal why vaccine coverage is low (National Vaccination Insights Project, 2025). A low coverage rate could be caused by several factors, including low consent rates, missed vaccination visits, vaccine-specific issues, or even changes in uptake over time. Without clearly separating these factors, it becomes increasingly difficult for state health department staff to know exactly where action may be required.

This project addressed that gap by developing a Power BI dashboard that goes beyond simply recording vaccination activity. The dashboard identifies schools below coverage thresholds, tracks consent rates, monitors yearly trends, and highlights operational issues such as incomplete visits. This aligns with the recommendations from the 2024 National Immunization strategy workshop, which highlighted the need for real-time vaccination coverage dashboards accessible to providers and stakeholders.

All in all, this project demonstrates how the state health department could move from spreadsheet-based reporting, towards a more proactive monitoring approach where low coverage and follow-up priorities can be identified not only earlier, but also more efficiently and effectively.

3. Data Exploration

3.1 Data in the System

The system was designed for use of vaccination data that is only available to the state health department and is far too granular to be publicly available. For the purposes of development, prototyping and demonstration, dummy data was generated using educated estimates. The state health department provided their current analytics format and reverse engineering. This showed the fields of the data that the system could expect to receive. There is a publicly available dataset containing information about public high schools with their name, suburb, local health district etc. The numbers we generated were assigned to a school's name, allowing for a one-to-many school to vaccination report relationship in the database and enabling further analysis such as coverage by location.

3.2 Data Generation

To generate the data, excel formulas were used to create biased random numbers for enrolments, consents, vaccinations, etc. Each being a weighted percentage of the previous as per the research indicating. Our research indicates that after the COVID-19 pandemic, the adolescent vaccine uptake was negatively impacted. (National Vaccination Insights Project, 2024). To highlight this, the consents were randomly chosen to be 70-90% of enrolments and actual vaccinations to be 70-90% of that number. As 70-90% is a wide range, the data was widely varied with some schools having a strong vaccination rate and others a poor rate. However, when generating the data, weightings and biases were not determined using the school's location or indigenous population which do show real world correlation and thus the dummy data is only for demonstration.

Vaccine	2021	2022	2023	Trend
dTpa (by age 15)	87.3%	86.9%	85.5%	Decline
Men ACWY (by age 17)	76.1%	75.9%	72.8%	Decline
Indigenous Men ACWY	66.7%	65.6%	62.3%	Decline

Source: National Vaccination Insights Project, 2024

3.3 Future Data Input

For future use of the system a Microsoft form (Appendix H) was created containing all the information that the nurses administering the vaccinations will be able to fill out. Each time the form is filled out the record will automatically be updated in the system.

4. Modelling

This section covers the modelling work conducted on the School Immunization Tracking System. In line with the CRISP-DM methodology adopted for this project, the modelling phase encompasses both the structural data model and the relational database schema designed to organize and validate immunization data. The analytical model, which applies data mining techniques to extract actionable insights for the state health department. The emphasis in this project is on a structural modelling approach, as defined in the original project proposal, where the relational schema and DAX-based measures in Power BI serve as the primary analytical engine rather than standalone algorithmic models.

4.1 Data Mining Problem

The business problem - the state health department's inability to efficiently monitor school vaccination coverage across 80 schools, 8 Local Health Districts and 3 vaccine types, was framed as a data mining problem where the goal is to extract meaningful, actionable insights from immunization data without requiring manual analysis. **4 data mining techniques were identified as relevant to this problem - Classification, Clustering, Association and Prediction.**

However, none of these techniques can operate reliably without a clean, structured and validated data foundation. The primary modelling challenge was therefore to **design and implement a relational database schema capable of storing, validating, and serving immunization data in a form that directly enables each of the 4 mining approaches**. The database model is not simply infrastructure but it encodes the analytical logic of the system.

Computed columns calculate coverage and consent rates automatically, reference tables embed herd immunity thresholds as model parameters and the staging-to-production pipeline enforces data quality as a precondition for reliable analysis.

The table below summarizes how each data mining technique maps to a business question, is enabled by the database model and outputs on the Power BI dashboard.

Technique	Business Question	How DB model enables it?	Dashboard Output
Classification	Which schools are below target coverage?	CoverageStatusRAG column stores Red/Amber/Green labels per record. VaccinationRateOfEnrolled computed column feeds the threshold logic.	RAG status donut chart and school-level status table on Overview page.
Clustering	Which schools share similar immunization patterns?	Region, IsSSP, Suburb, Postcode fields in Schools table enable grouping by geography and school type.	Coverage by Local Health District bar chart, geographic postcode map.
Association	Does Consent Rate predict Vaccination Coverage?	ConsentRate and VaccinationRateOfEnrolled stored as computed columns in the same record, enabling direct regression in Power BI.	Scatterplot showing r = 0.91 correlation between Consent Rate and Coverage Rate.
Prediction	Is coverage improving or declining over time?	ReportingYear field across 2024 - 2026 records allows multi-year trend aggregation.	Multi-Year Analytics page with year-on-year percentage change KPI cards.

Table 4.1: Data Mining Technique Mapping to Business Problem.

4.2 Relational Database Schema

The core of the analytical model is a normalised relational database hosted on **Microsoft Azure SQL (ImmunisationDB).** The schema was designed to replace the flat, disconnected Excel spreadsheets currently used by the state health department with a structured, validated and queryable data store. The schema consists of five tables, each serving a distinct role in the analytical pipeline.

Table	Primary Key	Key Fields	Details
Schools	SchoolID (INT, IDENTITY)	SchoolName, IsSSP, CreatedDate, Suburb, Postcode	Reference table for all 80 NSW schools grouped by Local Health District (LHD).
VaccineTypes	VaccineID (INT, IDENTITY)	VaccineID, VaccineName, TargetYearGroup, HerdImmunityThreshold	Stores the 3 vaccines (HPV, dTpa, Men ACWY) with WHO herd immunity thresholds as embedded model parameters.
VaccinationRecords	RecordID (INT, IDENTITY)	RecordID, SchoolID, VaccineID, VisitNumber, VisitDate, YearGroup, TotalEnrolled, TotalConsented, TotalVaccinated, VaccinatedFromPaperConsents, SubmissionDate, ConsentRate, VaccinationRateOfEnrolled, VaccinationRateOfConsented, ReportingYear, Email, PlannedVisits, VisitsCompleted, CatchUpMinExpected, CatchUpMaxExpected, ElectronicConsentVaccinated, VisitStatus, CoverageStatusRAG	Central fact table, stores per visit submission records with all raw counts, computed metrics and operational fields.
StagingTable	StagingID (INT, IDENTITY)	StagingID, RawSchoolName, RawVaccineType, RawVisitDate, RawVisitNumber, RawYearGroup, RawTotalEnrolled, RawTotalConsented, RawTotalVaccinated, RawPaperConsents, IsProcessed, ReceivedDate	Temporary holding table for raw Power Automate submissions before validation and promotion.
LHD_UserMapping	Composite (UserEmail, LHDName)	UserEmail, LHDName	Maps each database user to their assigned Local Health District, authoritative source for Row-Level Security enforcement.

The schema follows a star-schema pattern, with VaccinationRecords as the central fact table and Schools and VaccineTypes as dimension tables. This structure was chosen to align with Power BI's data model, where **fact-dimension relationships support efficient DAX aggregations and cross-filtering across dashboard pages**. The screenshot below shows the **production tables populated with live data in SQL Server Management Studio (SSMS),** confirming the Schools, VaccinationRecords and VaccineTypes tables were successfully created and loaded.

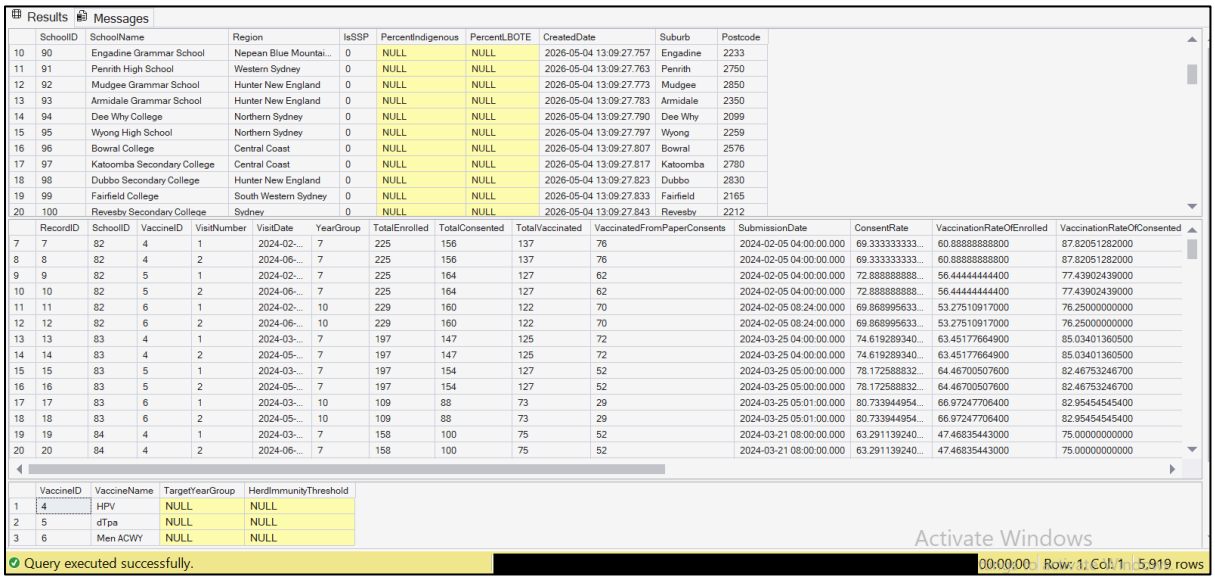


Figure 4.2: SSMS query results showing the Schools, VaccinationRecords and VaccineTypes production tables populated with live data.

A key design decision was to store herd immunity thresholds directly in the VaccineTypes reference table, sourced from WHO immunization guidelines and the Australian Immunization Handbook (2024). In the current prototype, the HerdImmunityThreshold field is NULL as these thresholds are applied within Power BI as RAG classification logic instead. In a production deployment, these values would be populated directly in the database. **Foreign key constraints between VaccinationRecords, Schools and VaccineTypes enforce referential integrity**. A **VisitNumber CHECK constraint (values 1, 2 or 3) limits visit entries to the program's defined schedule, preventing erroneous data from entering the production tables**.

4.3 Computed Columns as Analytical Measures

A significant modelling decision was the use of SQL computed columns within the VaccinationRecords table to derive key analytical metrics automatically at the database level. This ensures **coverage and consent rates are calculated consistently from stored raw counts rather than relying on potentially divergent manual calculations across different dashboard pages**.

Computed Column	SQL Formula	Analytical Purpose
ConsentRate	CAST(TotalConsented AS DECIMAL) / NULLIF(TotalEnrolled, 0) x 100	Parental consent uptake per visit, strongest relationship in the synthetic dataset (r = 0.91).
VaccinatonRateOfEnrolled	CAST(TotalVaccinated AS DECIMAL) /NULLIF(TotalEnrolled, 0) x 100	Primary coverage KPI, compared against the 80% target and vaccine-specific herd immunity thresholds.
VaccinatonRateOfConsented	CAST(TotalVaccinated AS DECIMAL) / NULLIF(TotalConsented, 0) x 100	Post-consent drop-off rate, identifies consented students who were not vaccinated and targets them for follow-up visits.

Table 4.3: Computed Columns and Formula in VaccinationRecords table.

From Table 4.3, the use of NULLIF(denominator, 0) in each formula prevents division-by-zero errors, **ensuring the model handles edge cases gracefully**. Notably, the ConsentRate column proved particularly significant, a **Pearson correlation of r = 0.91 between consent rate and vaccination coverage rate was observed in the synthetic dataset (where this relationship was built into the data by design), showing that the model can surface a consent-driven pattern,** directly answering the association mining question.

4.4 Staging Table and Stored Procedure

A 2-stage data pipeline was implemented to ensure that **raw form submissions are preserved and validated before being promoted to the production tables that Power BI queries.** This design choice was driven by the need to **protect data integrity in an environment where submissions arrive asynchronously** via Microsoft Forms and Power Automate.

4.4.1 Staging Table

All incoming data from Power Automate is inserted into a StagingTable first. The staging table stores all fields as VARCHAR types, accepting raw data without type enforcement. This acts as a safety buffer - if a form submission contains an invalid date format or a non-numeric enrolment count, the error is contained in staging rather than corrupting the production VaccinationRecords table. An IsProcessed BIT column (default 0) tracks which rows have been promoted, allowing the pipeline to be re-run safely without reprocessing completed records.

4.4.2 Stored Procedure (usp_ProcessStagingData)

The promotion logic is encapsulated in the stored procedure usp_ProcessStagingData, which is auto-triggered by Power Automate after each new staging insertion. The procedure performs the following operations in sequence within a single transaction:

1. Upserts new school entries into the Schools table if the school name does not already exist.
2. Upserts new vaccine type entries into VaccineTypes if the vaccine name is not already present.
3. Casts and inserts staged rows into VaccinationRecords with proper data types (INT for counts, DATE for visit dates), joining to Schools and VaccineTypes to resolve foreign key references.

4. Marks processed rows in StagingTable by setting IsProcessed = 1.

The stored procedure centralises all transformation logic in the database, meaning if the **logic needs to change, it is updated in one place rather than across multiple Power Automate flows or Python scripts.** Critically, the procedure provides transactional integrity. If any step fails, the entire operation rolls back cleanly, preventing partial or corrupt data from entering the production tables.

```
CREATE PROCEDURE usp_ProcessStagingData
AS
BEGIN
    -- Insert new schools if they don't exist
    INSERT INTO Schools (SchoolName, Region, IsSSP, CreatedDate)
    SELECT DISTINCT
        RawSchoolName,
        'Unknown',
        0,
        GETDATE()
    FROM StagingTable
    WHERE IsProcessed = 0
    AND RawSchoolName NOT IN (SELECT SchoolName FROM Schools);

    -- Insert new vaccine types if they don't exist
    INSERT INTO VaccineTypes (VaccineName)
    SELECT DISTINCT RawVaccineType
    FROM StagingTable
    WHERE IsProcessed = 0
    AND RawVaccineType NOT IN (SELECT VaccineName FROM VaccineTypes);

    -- Promote staged rows into VaccinationRecords
    INSERT INTO VaccinationRecords (
        SchoolID,
        VaccineID,
        VisitNumber,
        VisitDate,
        YearGroup,
        TotalEnrolled,
        TotalConsented,
        TotalVaccinated,
        VaccinatedFromPaperConsents
    )
    SELECT
        s.SchoolID,
        v.VaccineID,
        CAST(st.RawVisitNumber AS INT),
        CAST(st.RawVisitDate AS DATE),
        st.RawYearGroup,
        CAST(st.RawTotalEnrolled AS INT),
        CAST(st.RawTotalConsented AS INT),
        CAST(st.RawTotalVaccinated AS INT),
        CAST(st.RawPaperConsents AS INT)
    FROM StagingTable st
    JOIN Schools s ON s.SchoolName = st.RawSchoolName
    JOIN VaccineTypes v ON v.VaccineName = st.RawVaccineType
    WHERE st.IsProcessed = 0;

    -- Mark staged rows as processed
    UPDATE StagingTable
    SET IsProcessed = 1
    WHERE IsProcessed = 0;
END;
GO
```

4.4.3 Python Data Integration Script

In addition to the Power Automate pipeline, **a Python script using the pyodbc library was developed to load the 720-row dummy dataset directly into the staging and production tables**. The script performs column renaming to align the CSV field names with the database schema, NULL handling for optional fields, and iterates over all three visit date columns per record to insert a separate staging row for each scheduled visit. This approach correctly models the per-visit structure of the program, where each school-vaccine-visit combination is an independent submission record.

The script successfully loaded 720 rows across all 3 vaccines and 3 reporting years (2024 -2026), which were subsequently promoted to VaccinationRecords via the stored procedure. This confirmed that the end-to-end pipeline was functional before connecting the live Power Automate flow.

The Python Script can be viewed below:

```
# School Immunisation Tracking System: synthetic data loader
# Credentials are read from environment variables

import pandas as pd
import pyodbc
import os

'''
Python will “talk” to the SQL server in the backend. So, it needs a way to communicate with the server &
thus it uses a Driver, which acts like a translator pyodbc is a library that connects Python to the SQL Server.

'''

conn = pyodbc.connect(
    'DRIVER={ODBC Driver 18 for SQL Server};'
    f'SERVER={os.environ["DB_SERVER"]};'
    f'DATABASE={os.environ["DB_NAME"]};'
    f'UID={os.environ["DB_USER"]};'
    f'PWD={os.environ["DB_PASSWORD"]};'
    'Encrypt=yes;'
    'TrustServerCertificate=no;'
    'Connection Timeout=30;'

)
cursor = conn.cursor()


# Load dummy dataset & specify the file path for dummy dataset.
df_dummy = pd.read_csv('data/dummy_data_v2.csv')

# Clean the column names (words with spaces).
df_dummy.columns = df_dummy.columns.str.strip().str.replace(' ', '')

# Rename the columns from the dummy dataset to match the SQL insert.
df_dummy.rename(columns={

    'SubmissionTime': 'SubmissionTime',
    'Email': 'Email',
    'Year': 'Year',
    'YearGroup': 'YearGroup',
    'VaccineBeingAdministered': 'VaccineType',

    'SchoolName': 'SchoolName',
    'Suburb' : 'Suburb',
    'Postcode' : 'Postcode',
```

```
    'LocalHealthDistrict' : 'LocalHealthDistrict',
    'IsSSPSchool' : 'SSPSchool?',

    'TotalEnrolled': 'TotalEnrolled',
    'Consented(PaperandElectronic)': 'Consented(Paper&Electronic)',
    'TotalVaccinated(IncludingPaperConsents)': 'TotalVaccinated',
    'NumberVaccinated(PaperConsentsOnly)': 'PaperConsents',
    'PlannedVisits': 'PlannedVisits',

    'VisitsCompleted' : 'VisitsCompleted',
    'Visit1Date': 'Visit1Date',
    'Visit2Date': 'Visit2Date',
    'Visit3Date': 'Visit3Date',
    'Catch-UpMinExpected(ConsentedNotVaccinated)': 'ConsentedNotVaccinated',

    'Catch-UpMaxExpected(EnrolledNotVaccinated)' : 'EnrolledNotVaccinated',
    '%Consented(ofEnrolled)' : '%Consented(OfEnrolled)',
    '%Vaccinated(ofEnrolled)' : '%Vaccinated(Enrolled)',
    '%Vaccinated(ofConsented)' : '%Vaccinated(Consented)',
    'NumberVaccinated(ElectronicConsentsOnly)': 'E-Consents',

    'CoverageStatus(RAG)' : 'Coverage Status(RAG)',
    'VisitStatus' : 'VisitStatus'

}, inplace=True)

# ---------------------------------------------------------------------------- Fix NULL handling to match actual
renamed columns ------------------------------------------------------------------
df_dummy = df_dummy.fillna({
    'SchoolName':    '',
    'VaccineType':   '',
    'Visit1Date':    '',
    'Visit2Date':    '',
    'Visit3Date':    '',
    'YearGroup':     0,
    'TotalEnrolled': 0,
    'Consented(Paper&Electronic)': 0,
    'TotalVaccinated': 0,
    'PaperConsents': 0,
    'E-Consents':    0,
    'PlannedVisits': 0,
    'VisitsCompleted': 0,
    'ConsentedNotVaccinated': 0,
    'EnrolledNotVaccinated':  0,
    '%Consented(OfEnrolled)':  0.0,
    '%Vaccinated(Enrolled)':   0.0,
    '%Vaccinated(Consented)':  0.0,
    'Suburb':              '',
```

```
    'LocalHealthDistrict': '',
    'Coverage Status(RAG)': '',
    'VisitStatus':          '',
})

print("COLUMNS:", df_dummy.columns.tolist())
print(df_dummy.head())


# ------------------------------------------------------------------------------ INSERT INTO STAGING TABLE
------------------------------------------------------------------------
# Iterates over all 3 visit dates so each visit gets its own staging row.
for _, row in df_dummy.iterrows():
    for visit_num, visit_col in enumerate(['Visit1Date', 'Visit2Date', 'Visit3Date'], start=1):
        visit_date = row[visit_col]
        if not visit_date: # Skip visits that don't exist for this record.
            continue

        cursor.execute("""
            INSERT INTO StagingTable (
                RawSchoolName,
                RawVaccineType,
                RawVisitDate,
                RawVisitNumber,
                RawYearGroup,
                RawTotalEnrolled,
                RawTotalConsented,
                RawTotalVaccinated,
                RawPaperConsents
            )
            VALUES (?, ?, ?, ?, ?, ?, ?, ?, ?)
        """,
            row['SchoolName'],
            row['VaccineType'],
            visit_date,
            visit_num,
            row['YearGroup'],
            row['TotalEnrolled'],
            row['Consented(Paper&Electronic)'],
            row['TotalVaccinated'],
            row['PaperConsents']
        )

conn.commit()
print("Staging table insert complete.")


# ------------------------------------------------------------------------------ INSERT INTO SCHOOLS TABLE -
------------------------------------------------------------------------
```

```
unique_schools = df_dummy[['SchoolName', 'LocalHealthDistrict', 'SSPSchool?', 'Suburb',
'Postcode']].drop_duplicates()

for _, school in unique_schools.iterrows():
    cursor.execute("""
    IF NOT EXISTS (SELECT 1 FROM Schools WHERE SchoolName = ?)
    INSERT INTO Schools (SchoolName, Region, IsSSP, Suburb, Postcode, CreatedDate)
    VALUES (?, ?, ?, ?, ?, GETDATE())
""",
    school['SchoolName'],
    school['SchoolName'],
    school['LocalHealthDistrict'],
    int(school['SSPSchool?']),
    school['Suburb'],
    school['Postcode']
)


conn.commit()
print("Schools insert complete.")


# ------------------------------------------------------------------------------ INSERT INTO VACCINETYPES
TABLE ------------------------------------------------------------------
unique_vaccines = df_dummy[['VaccineType']].drop_duplicates()

for _, vax in unique_vaccines.iterrows():
    cursor.execute("""
        IF NOT EXISTS (SELECT 1 FROM VaccineTypes WHERE VaccineName = ?)
        INSERT INTO VaccineTypes (VaccineName)
        VALUES (?)
    """,
        vax['VaccineType'],
        vax['VaccineType']
    )

conn.commit()
print("VaccineTypes insert complete.")

# ------------------------------------------------------------------------------ INSERT INTO
VACCINATIONRECORDS ------------------------------------------------------------------------------
for _, row in df_dummy.iterrows():
    for visit_num, visit_col in enumerate(['Visit1Date', 'Visit2Date', 'Visit3Date'], start=1):
        visit_date = row[visit_col]
        if not visit_date:
            continue

        cursor.execute("""
```

```
    INSERT INTO VaccinationRecords (
        SchoolID,
        VaccineID,
        VisitNumber,
        VisitDate,
        YearGroup,
        TotalEnrolled,
        TotalConsented,
        TotalVaccinated,
        VaccinatedFromPaperConsents,
        SubmissionDate,
        ReportingYear,
        Email,
        PlannedVisits,
        VisitsCompleted,
        CatchUpMinExpected,
        CatchUpMaxExpected,
        ElectronicConsentVaccinated,
        VisitStatus,
        CoverageStatusRAG
    )
    SELECT
        s.SchoolID,
        v.VaccineID,
        ?, ?, ?, ?, ?, ?, ?, ?, ?, ?, ?, ?, ?, ?, ?, ?, ?
    FROM Schools s
    JOIN VaccineTypes v ON v.VaccineName = ?
    WHERE s.SchoolName = ?
""",
    visit_num,
    visit_date,
    row['YearGroup'],
    row['TotalEnrolled'],
    row['Consented(Paper&Electronic)'],
    row['TotalVaccinated'],
    row['PaperConsents'],
    row['SubmissionTime'],

    row['Year'],
    row['Email'],
    row['PlannedVisits'],
    row['VisitsCompleted'],
    row['ConsentedNotVaccinated'],
    row['EnrolledNotVaccinated'],
    row['E-Consents'],
    row['VisitStatus'],
    row['Coverage Status(RAG)'],
```

```
    # For the JOIN
    row['VaccineType'],
    row['SchoolName']
)

conn.commit()
print("VaccinationRecords insert complete.")
```

```
0   1  2024-03-25 05:00:00  ...    Red     Complete
1   2  2024-03-25 05:00:00  ...  Amber     Complete
2   3  2024-03-25 08:37:00  ...  Amber     Complete
3   4  2024-02-05 04:00:00  ...  Amber     Complete
4   5  2024-02-05 04:00:00  ...    Red     Complete

[5 rows x 28 columns]
Staging table insert complete.
Schools insert complete.
VaccineTypes insert complete.
VaccinationRecords insert complete.

[Done] exited with code=0 in 19.216 seconds
```

Figure 4.3: Terminal output confirming successful completion of table inserts.

4.5 Row-Level Security Model

A Row-Level Security (RLS) access control model was implemented at the database level to enforce data governance requirements. The state health department operates across 8 Local Health Districts, each managed by a separate coordinator. It is a governance requirement that coordinators can only **view data for schools within their own district** because **cross-district data access would create privacy and operational risks**.

The RLS model was implemented using the following components:

1. **LHD_UserMapping table:** Stores the mapping between each database user's email/username and their assigned Local Health District. This serves as the authoritative source for access boundaries.

2. **Security schema and filter function (Security.fn_LHD_Filter)**: An inline table-valued function that evaluates 2 conditions using an OR operator: (1) if the current user is a db_owner (admin), all rows are returned (2). Otherwise, only rows where the Region column matches the user's assigned LHD from LHD_UserMapping are returned.

3. **Security policy (LHD_SchoolPolicy):** Applied as a FILTER PREDICATE on the Schools table, which propagates through the VaccinationRecords join relationships, ensuring coordinators can only reach records for their district's schools.

Since the real staff email addresses and usernames were not available during development, **placeholder coordinator accounts were created for prototype**. These dummy accounts were inserted into the LHD_UserMapping table to represent each of the 8 Local Health Districts. In addition, the team's own database usernames (test_user_1, test_user_2, test_user_3) were mapped to specific LHDs to **enable live impersonation testing of the RLS policy within SSMS**. In a **production deployment, these placeholder entries would be replaced with verified state health department staff credentials** and the LHD_UserMapping table would be updated accordingly without any changes to the security policy or filter function itself. The steps to implement RLS is as follows:

1. Create a table to store the Emails and Region values.

```
CREATE TABLE LHD_UserMapping (
UserEmail VARCHAR(100),
LHDName VARCHAR(100)

)
```

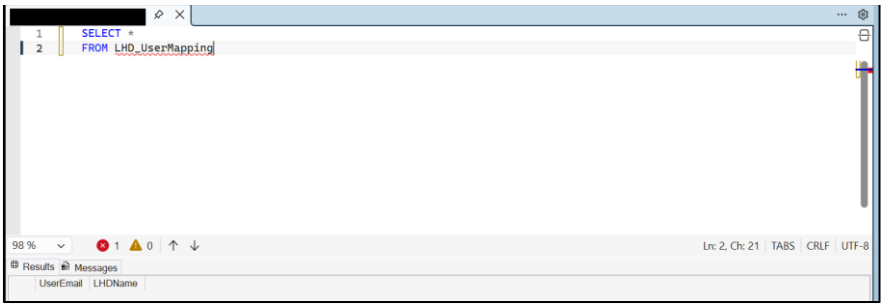


Figure 4.5.1: UserEmail and LHDName columns are created.

2. Create some fake entries.

```
INSERT INTO LHD_UserMapping VALUES
('coordinator.westernsydney@example.test', 'Western Sydney'),
('coordinator.sydney@example.test', 'Sydney'),
('coordinator.northernsydney@example.test', 'Northern Sydney'),
('coordinator.southeastsydney@example.test', 'South Eastern Sydney'),
('coordinator.southwestsydney@example.test', 'South Western Sydney'),
('coordinator.nepean@example.test', 'Nepean Blue Mountains'),
('coordinator.hunter@example.test', 'Hunter New England'),
('coordinator.centralcoast@example.test', 'Central Coast');
```

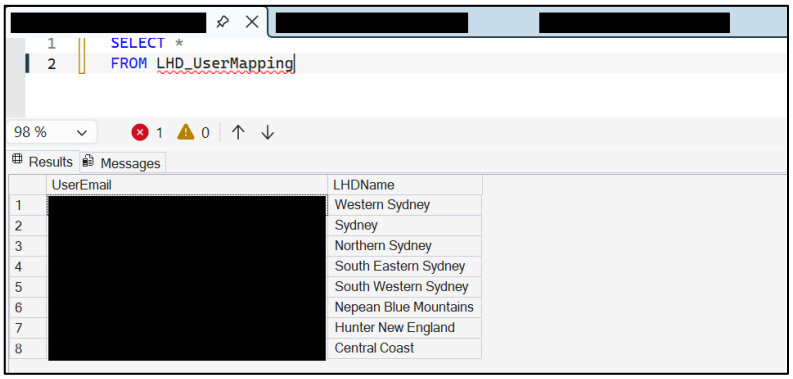


Figure 4.5.2 : LHD_UserMapping table showing all 8 regions mapped to the dummy email accounts.

3. Enable filtering function based on 2 conditions, using OR operator.

```
-- Create a separate schema to store security-related objects.
CREATE SCHEMA Security;
GO

-- Create a Row-Level Security filter function.
CREATE FUNCTION Security.fn_LHD_Filter(@Region NVARCHAR(100))
RETURNS TABLE
WITH SCHEMABINDING
AS
RETURN
(
```

```
    SELECT 1 AS fn_result
    WHERE
        -- Condition 1: If the user is a database owner, allow access to all rows.
        IS_MEMBER('db_owner') = 1

        OR

        -- Condition 2: Allow access only if the row's region matches.
        -- the region assigned to the current logged-in user.
        @Region IN (
            SELECT LHDName
            FROM dbo.LHD_UserMapping
            WHERE UserEmail = USER_NAME()  -- Gets the current user's login.
        )
);

GO
```

4. Apply security policy to Schools table.

```
CREATE SECURITY POLICY LHD_SchoolPolicy
ADD FILTER PREDICATE Security.fn_LHD_Filter(Region)
ON dbo.Schools
WITH (STATE = ON);
```

5. Map entries to team usernames & test to see if filter has been applied.

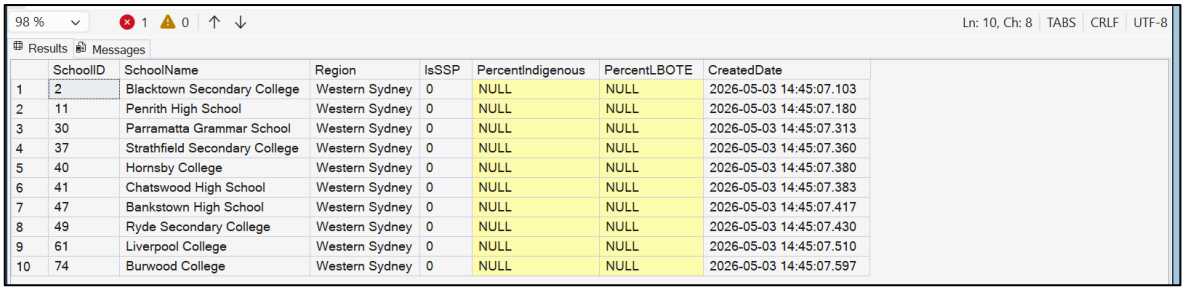
```
-- Temporarily map team's DB users to LHDs for testing.
INSERT INTO LHD_UserMapping VALUES
('test_user_1', 'Western Sydney'),
('test_user_2', 'Sydney'),
('test_user_3', 'Northern Sydney');

-- Test
EXECUTE AS USER = 'test_user_1';
SELECT * FROM Schools;  -- should only return Western Sydney schools
REVERT;
```

Results | Messages

	UserEmail	LHDName
1	[redacted]	Western Sydney
2		Sydney
3		Northern Sydney
4		South Eastern Sydney
5		South Western Sydney
6		Nepean Blue Mountains
7		Hunter New England
8		Central Coast
9		Western Sydney
10		Sydney
11		Northern Sydney
12		Western Sydney
13		Sydney
14		Northern Sydney
15		Western Sydney
16		Sydney
17		Northern Sydney
18		Western Sydney
19		Sydney
20		Northern Sydney
21		Western Sydney
22		Sydney
23		Northern Sydney

Figure 4.5.3: Confirm mapping User to Region is done.

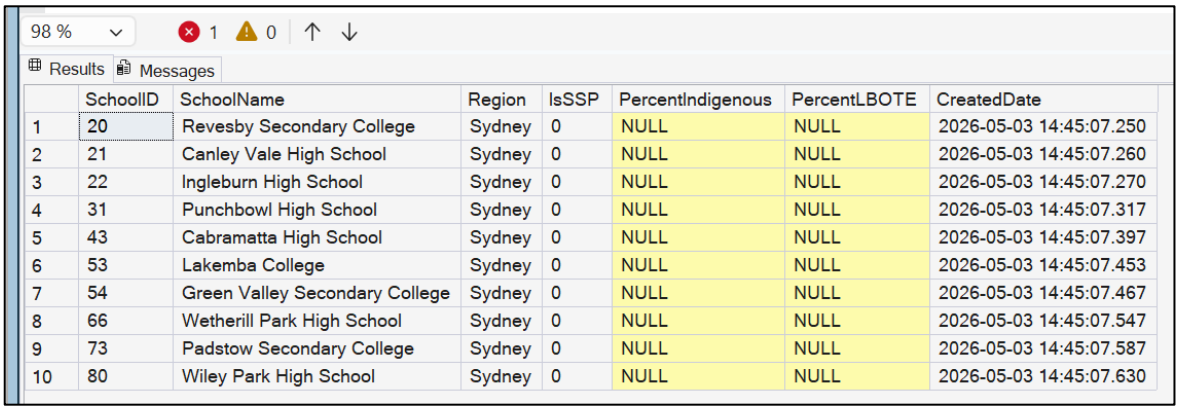


98 % 1 0 Ln: 10, Ch: 8 TABS CRLF UTF-8

Results Messages

	SchoolID	SchoolName	Region	IsSSP	PercentIndigenous	PercentLBOTE	CreatedDate
1	2	Blacktown Secondary College	Western Sydney	0	NULL	NULL	2026-05-03 14:45:07.103
2	11	Penrith High School	Western Sydney	0	NULL	NULL	2026-05-03 14:45:07.180
3	30	Parramatta Grammar School	Western Sydney	0	NULL	NULL	2026-05-03 14:45:07.313
4	37	Strathfield Secondary College	Western Sydney	0	NULL	NULL	2026-05-03 14:45:07.360
5	40	Hornsby College	Western Sydney	0	NULL	NULL	2026-05-03 14:45:07.380
6	41	Chatswood High School	Western Sydney	0	NULL	NULL	2026-05-03 14:45:07.383
7	47	Bankstown High School	Western Sydney	0	NULL	NULL	2026-05-03 14:45:07.417
8	49	Ryde Secondary College	Western Sydney	0	NULL	NULL	2026-05-03 14:45:07.430
9	61	Liverpool College	Western Sydney	0	NULL	NULL	2026-05-03 14:45:07.510
10	74	Burwood College	Western Sydney	0	NULL	NULL	2026-05-03 14:45:07.597

Figure 4.5.4: Test user 1's DB view, only Western Sydney records.

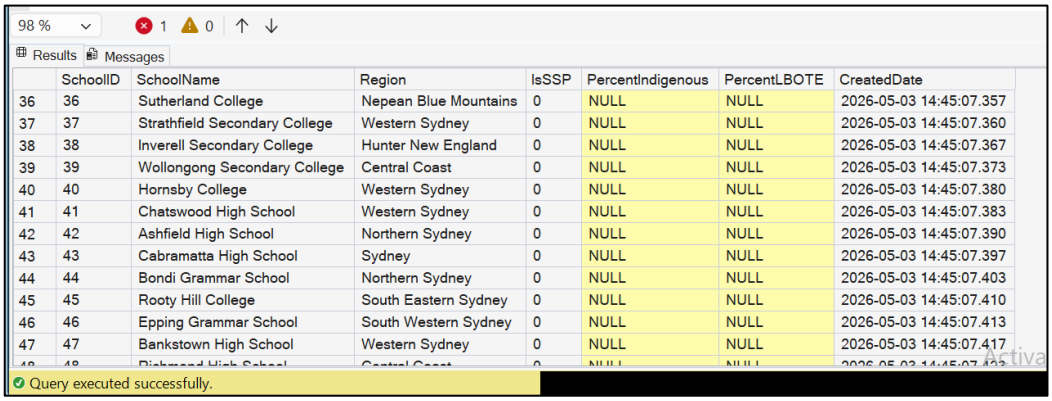


98 % 1 0

Results Messages

	SchoolID	SchoolName	Region	IsSSP	PercentIndigenous	PercentLBOTE	CreatedDate
1	20	Revesby Secondary College	Sydney	0	NULL	NULL	2026-05-03 14:45:07.250
2	21	Canley Vale High School	Sydney	0	NULL	NULL	2026-05-03 14:45:07.260
3	22	Ingleburn High School	Sydney	0	NULL	NULL	2026-05-03 14:45:07.270
4	31	Punchbowl High School	Sydney	0	NULL	NULL	2026-05-03 14:45:07.317
5	43	Cabramatta High School	Sydney	0	NULL	NULL	2026-05-03 14:45:07.397
6	53	Lakemba College	Sydney	0	NULL	NULL	2026-05-03 14:45:07.453
7	54	Green Valley Secondary College	Sydney	0	NULL	NULL	2026-05-03 14:45:07.467
8	66	Wetherill Park High School	Sydney	0	NULL	NULL	2026-05-03 14:45:07.547
9	73	Padstow Secondary College	Sydney	0	NULL	NULL	2026-05-03 14:45:07.587
10	80	Wiley Park High School	Sydney	0	NULL	NULL	2026-05-03 14:45:07.630

Figure 4.5.5: Test user 2's DB view, only Sydney records.



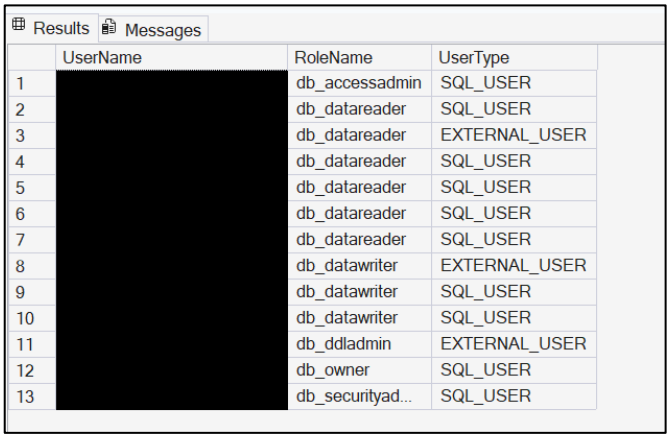
98 % 1 0

Results Messages

	SchoolID	SchoolName	Region	IsSSP	PercentIndigenous	PercentLBOTE	CreatedDate
36	36	Sutherland College	Nepean Blue Mountains	0	NULL	NULL	2026-05-03 14:45:07.357
37	37	Strathfield Secondary College	Western Sydney	0	NULL	NULL	2026-05-03 14:45:07.360
38	38	Inverell Secondary College	Hunter New England	0	NULL	NULL	2026-05-03 14:45:07.367
39	39	Wollongong Secondary College	Central Coast	0	NULL	NULL	2026-05-03 14:45:07.373
40	40	Hornsby College	Western Sydney	0	NULL	NULL	2026-05-03 14:45:07.380
41	41	Chatswood High School	Western Sydney	0	NULL	NULL	2026-05-03 14:45:07.383
42	42	Ashfield High School	Northern Sydney	0	NULL	NULL	2026-05-03 14:45:07.390
43	43	Cabramatta High School	Sydney	0	NULL	NULL	2026-05-03 14:45:07.397
44	44	Bondi Grammar School	Northern Sydney	0	NULL	NULL	2026-05-03 14:45:07.403
45	45	Rooty Hill College	South Eastern Sydney	0	NULL	NULL	2026-05-03 14:45:07.410
46	46	Epping Grammar School	South Western Sydney	0	NULL	NULL	2026-05-03 14:45:07.413
47	47	Bankstown High School	Western Sydney	0	NULL	NULL	2026-05-03 14:45:07.417

Query executed successfully.

Figure 4.5.6: All records for DB admin.



Results Messages

	UserName	RoleName	UserType
1	[redacted]	db_accessadmin	SQL_USER
2		db_datareader	SQL_USER
3		db_datareader	EXTERNAL_USER
4		db_datareader	SQL_USER
5		db_datareader	SQL_USER
6		db_datareader	SQL_USER
7		db_datareader	SQL_USER
8		db_datawriter	EXTERNAL_USER
9		db_datawriter	SQL_USER
10		db_datawriter	SQL_USER
11		db_ddladmin	EXTERNAL_USER
12		db_owner	SQL_USER
13		db_securityad...	SQL_USER

Figure 4.5.7: SSMS query result showing all roles/permissions across team members.

The Azure firewall was configured to allow all team members' IP addresses plus the *'Allow Azure services and resources to access this server'* exception, enabling the Power BI Service to connect directly to Azure SQL for scheduled dashboard refreshes.

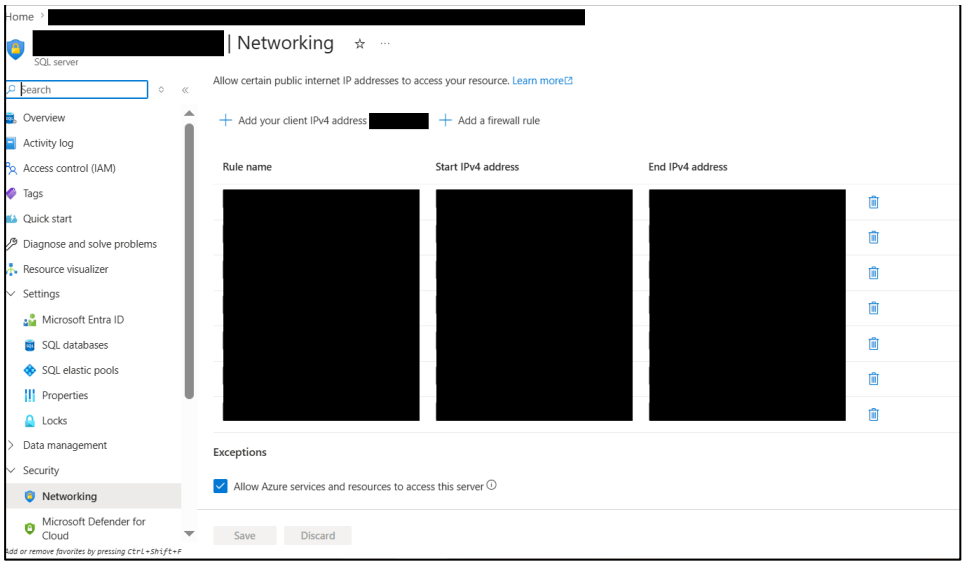


Figure 4.5.8: Azure SQL Server Networking Configuration.

4.6 Validation Audit

A validation audit was conducted across 5 dimensions to verify data integrity, model correctness and security enforcement before connecting the database to the Power BI dashboard. This audit constitutes the model quality assessment for this project.

4.6.1 Check 1: Staging Table Ingestion

A **COUNT(*)** query confirmed total rows received, followed by NULL checks on all for 4 critical fields (SchoolName, VaccineType, TotalEnrolled, TotalVaccinated). Figure 4.6.1.2 below confirmed the **14,940 staged rows with 0 NULLs in any critical field. (Note: the NullSchoolName column in the screenshot was produced with COUNT(*), so it shows the total row count rather than a NULL count; the corrected expression is SUM(CASE WHEN RawSchoolName IS NULL THEN 1 ELSE 0 END).)**

```
-- Total rows received
SELECT COUNT(*) AS TotalStagedRows FROM StagingTable;

-- Check for critical nulls
SELECT
    COUNT(*) AS NullSchoolName,
    SUM(CASE WHEN RawVaccineType IS NULL THEN 1 ELSE 0 END) AS NullVaccineType,
    SUM(CASE WHEN RawTotalEnrolled IS NULL THEN 1 ELSE 0 END) AS NullEnrolled,
    SUM(CASE WHEN RawTotalVaccinated IS NULL THEN 1 ELSE 0 END) AS NullVaccinated

FROM StagingTable;
```

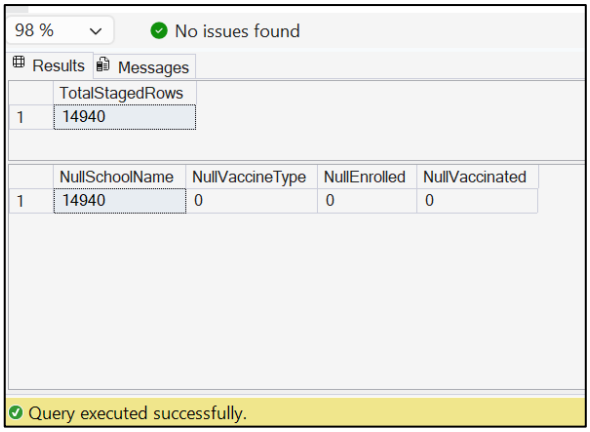


Figure 4.6.1.2: Staging Table Ingestion Audit.

4.6.2 Check 2: Logical Integrity

3 integrity checks were run to confirm no impossible values existed in VaccinationRecords - vaccinations exceeding enrolments, consents exceeding enrolments, and vaccinations exceeding consents. All three returned 0, confirming the dataset contains no logically impossible records.

```
-- Vaccinated should never exceed enrolled.
SELECT COUNT(*) AS ImpossibleVaccinationCount
FROM VaccinationRecords
WHERE TotalVaccinated > TotalEnrolled;

-- Consented should never exceed enrolled.
SELECT COUNT(*) AS ImpossibleConsentCount
FROM VaccinationRecords
WHERE TotalConsented > TotalEnrolled;

-- Vaccinated should never exceed consented.
SELECT COUNT(*) AS VaccinatedExceedsConsented
FROM VaccinationRecords
WHERE TotalVaccinated > TotalConsented;
```

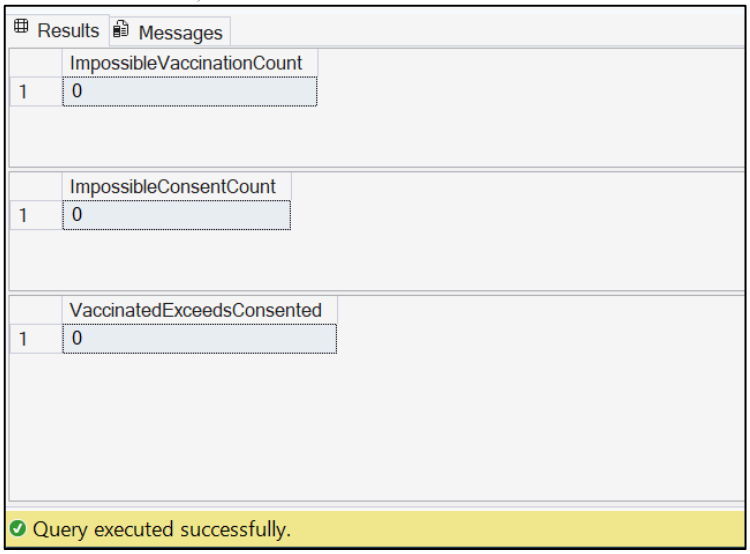


Figure 4.6.2.1: Logical Integrity Audit, confirming no logically invalid records exist in VaccinationRecords table.

4.6.3 Check 3: Coverage Rate Sanity

MIN, MAX and AVG were calculated on the VaccinationRateOfEnrolled computed column across all VaccinationRecords. The average returned ~ 53.2%, consistent with the **53.2% mean coverage displayed on the Power BI KPI cards**. This **cross-validated that the computed columns and Power BI DAX measures produce consistent outputs from the same underlying data.**

```
SELECT
    MIN(VaccinationRateOfEnrolled) AS MinCoverage,
    MAX(VaccinationRateOfEnrolled) AS MaxCoverage,
    AVG(VaccinationRateOfEnrolled) AS AvgCoverage
FROM VaccinationRecords;
```

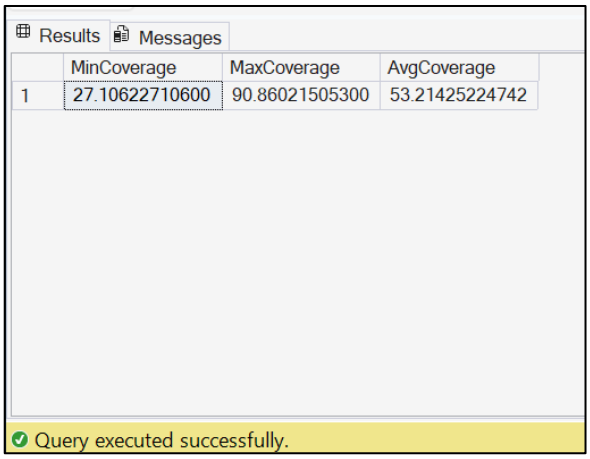


Figure 4.6.2.2: Coverage Rate sanity check - consistent with Power BI dashboard KPI, confirming computed column logic is correct.

4.6.4 Check 4: Row-Level Security Enforcement

To quantitatively verify the RLS policy, a count test was executed comparing the number of schools visible to an admin versus a coordinator. The results confirmed the policy is functioning as intended - the **admin account returned all 80 schools across every LHD, while the coordinator account was restricted to only 10 schools within their assigned district**.

```
-- Admin View: Should see all 80 schools
SELECT COUNT(*) AS AdminCanSee FROM Schools;

-- Coordinator View: Should see only their LHD/Region
EXECUTE AS USER = 'test_user_1';
SELECT COUNT(*) AS CoordinatorCanSee FROM Schools;
REVERT;
```

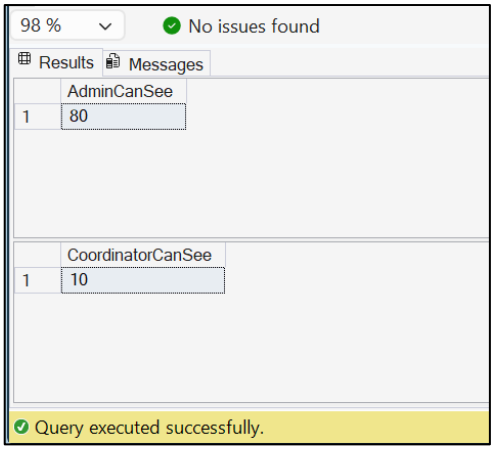


Figure 4.6.4.1: RLS count test confirming admin and coordinator data access boundaries.

4.6.5 Check 5: Duplicate Detection and Remediation

A **GROUP BY** query on RawSchoolName, RawVaccineType, and RawVisitDate with **HAVING COUNT(*) > 1** was run to **identify duplicate entries in the staging table. Duplicates were found as a result of the Python ingestion script being executed multiple times** during development.

```
SELECT
    RawSchoolName,
    RawVaccineType,
    RawVisitDate,
    COUNT(*) AS DuplicateCount
FROM StagingTable
GROUP BY RawSchoolName, RawVaccineType, RawVisitDate
HAVING COUNT(*) > 1;
```

Results Messages

	RawSchoolName	RawVaccineType	RawVisitDate	DuplicateCount
1				1440
2	Armidale Grammar School	dTpa	2024-02-21	11
3	Armidale Grammar School	dTpa	2024-06-06	6
4	Armidale Grammar School	dTpa	2025-02-13	11
5	Armidale Grammar School	dTpa	2025-05-28	6
6	Armidale Grammar School	dTpa	2026-03-20	11
7	Armidale Grammar School	dTpa	2026-06-01	6
8	Armidale Grammar School	HPV	2024-02-21	11
9	Armidale Grammar School	HPV	2024-06-06	6
10	Armidale Grammar School	HPV	2025-02-13	11
11	Armidale Grammar School	HPV	2025-05-28	6
12	Armidale Grammar School	HPV	2026-03-20	11
13	Armidale Grammar School	HPV	2026-06-01	6
14	Armidale Grammar School	Men ACWY	2024-02-21	11
15	Armidale Grammar School	Men ACWY	2024-06-06	6
16	Armidale Grammar School	Men ACWY	2025-02-13	11
17	Armidale Grammar School	Men ACWY	2025-05-28	6
18	Armidale Grammar School	Men ACWY	2026-03-20	11
19	Armidale Grammar School	Men ACWY	2026-06-01	6
20	Ashfield High School	dTpa	2024-02-26	11
21	Ashfield High School	dTpa	2024-06-21	6

Query executed successfully.

Figure 4.6.5.1: Duplicate detection results – multiple duplicate entries found.

The **staging table was truncated and the Python script was run once in a controlled single execution.** The duplicate check was then re-run, **returning an empty result set confirming complete remediation**.

```
SELECT COUNT(*) AS TotalRows FROM StagingTable;
```

```
-- Remove all duplicates by truncating.
TRUNCATE TABLE StagingTable;
```

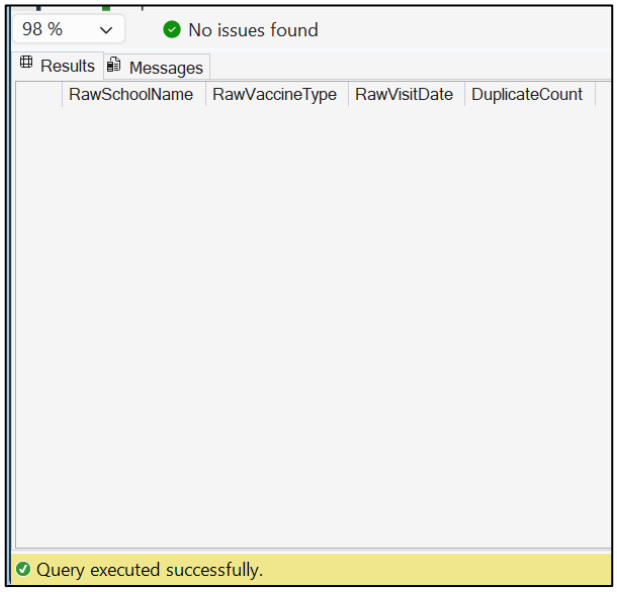


Figure 4.6.5.2 – Duplicate check after remediation.

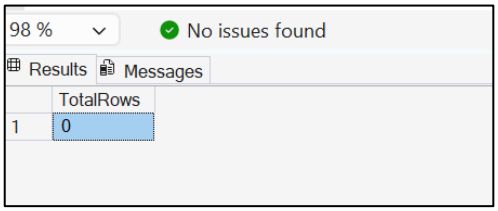


Figure 4.6.5.3: StagingTable results after TRUNCATE.

This finding highlighted the **importance of implementing upsert logic (checking for existing records before insertion)** in the production pipeline to prevent duplication when **Power Automate re-triggers on the same form submission.** This has been documented as a future improvement in the deployment plan.

4.7 Live Data Integration and Refresh Model

For the Power BI dashboard to function as a live monitoring tool, the database must support scheduled refresh connections from the Power BI Service. The following configuration was implemented:

• **Azure Firewall Rule**: The *'Allow Azure services and resources to access this server'* rule was enabled, permitting Power BI Service to connect directly to ImmunisationDB without a gateway or VPN.

• **Read Access for Refresh**: All reporting users were assigned the db_datareader role, granting read access to production tables without elevated write privileges.

• **Refresh Schedule:** The database is configured to support a scheduled refresh cycle, meaning the dashboard reflects new submissions within this window after staging promotion.

• **Staging Table Isolation:** Power BI queries VaccinationRecords exclusively, not StagingTable. Submissions arriving during a refresh window are safely held in staging until the stored procedure promotes them, preventing incomplete records from appearing in live reports.

A sanity check query was executed against VaccinationRecords to **confirm record count and date range before publishing the dashboard.**

```sql
-- Confirm VaccinationRecords has data ready for Power BI to read.
SELECT
    COUNT(*) AS TotalRecords,
    MIN(SubmissionDate) AS EarliestRecord,
    MAX(SubmissionDate) AS LatestRecord
FROM VaccinationRecords;
```

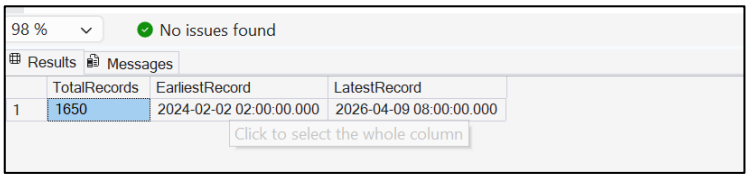


Figure 4.7.1: VaccinationRecords sanity check - 1,650 total records fully populated and ready for Power BI connection.

5. Data Exploration

To demonstrate the capability of the analytical model before the live pipeline is operational, we intentionally generated a dataset of 80 NSW schools across 8 Local Health Districts, 3 vaccine types (*HPV, dTpa, Men ACWY*), and 3 reporting years (*2024-2026*), with characteristics built in: *coverage rates* set below herd immunity thresholds, *consent gaps*, *geographic variation* between districts, and a simulated *post COVID coverage dip*. Each of these was a deliberate decision to verify that the model correctly detects, surfaces, and communicates these issues with designed data, giving us confidence it will do the same when real data flows through.

The analysis begins with an overview of program performance and explores the distribution of vaccination rates by school, major factors affecting vaccination participation, geographic variation across Local Health Districts, trends over time, and potential for future improvement. The core value of the system we are building for the state health department is that any coordinator, LHD manager, or school nurse can open the dashboard, identify these issues in real time, and flag them without needing to request a report or interpret a spreadsheet.

5.1 Overall Program Performance

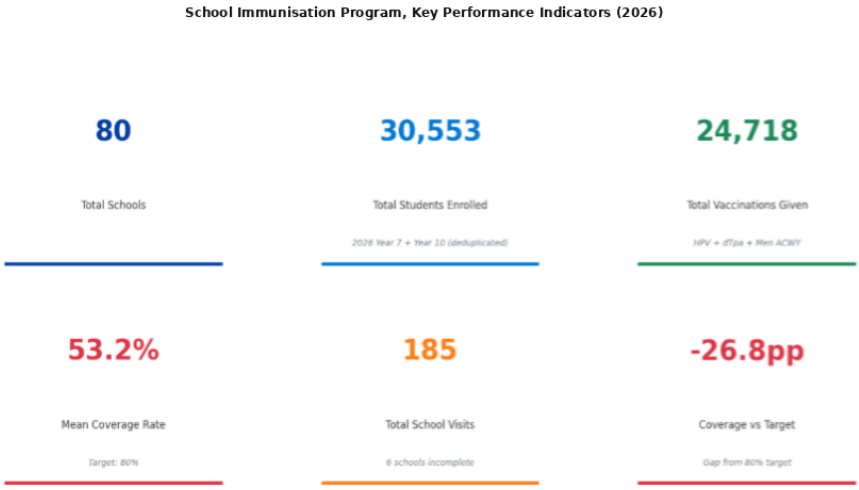


Figure 1. Program KPI Overview

Figure 1 summarises the overall size and key performance indicators of the NSW school vaccination program. A total of 80 schools are participating in the program, and 30,553 students are enrolled. A total of 24,718 vaccinations were administered, and the overall average vaccination rate was 53.2%. This is approximately 26.8% lower than the generally recommended 80% target vaccination rate in public health programs.

Even though vaccination programs are being operated for many schools and students, the actual vaccination participation rate does not meet expectations. Furthermore, these overall average figures do not reflect differences in inoculation rates by school. It suggests that a more detailed analysis at the school level is necessary to understand the actual program performance. We deliberately set coverage to fall in this range to test whether the KPI cards would surface this deficit clearly and immediately, without any manual aggregation. They did, confirming that any coordinator opening the live dashboard can answer "are we on track?" in seconds rather than collating spreadsheets across districts.

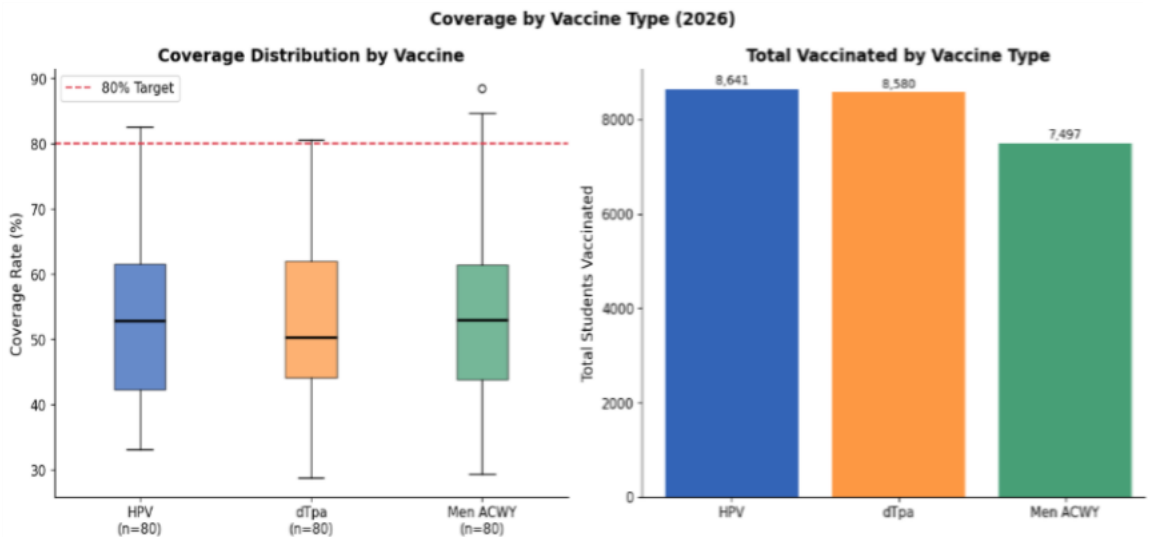


Figure 2. Vaccination Coverage by Vaccine Type

Figure 2 shows the results of comparing the average vaccination rates for the three vaccines (HPV, dTpa, and Men ACWY). HPV had an average vaccination rate of 53.3%, dTpa had an average vaccination rate of 52.7%, and Men ACWY had an average vaccination rate of 53.5%, with little difference between the three vaccines. The similar vaccination rates for all vaccines at the level of about 52 to 54% suggests the possibility of a structural problem with lower student participation rates across the program, rather than a problem with specific vaccines. This means that actual vaccination participation remains at half the level, even though vaccination programs are operating in the same way in several schools, suggesting that factors such as parental consent rates, student absences, or awareness of vaccination participation may affect vaccination rates.

We deliberately set all three vaccines to similar coverage ranges to test whether the model would distinguish between a vaccine specific problem and a systemic one. It correctly did, meaning a coordinator can filter by vaccine type on the live dashboard and reach the same conclusion in seconds, the root cause is upstream of the vaccine itself.

5.2 Vaccination Coverage Patterns

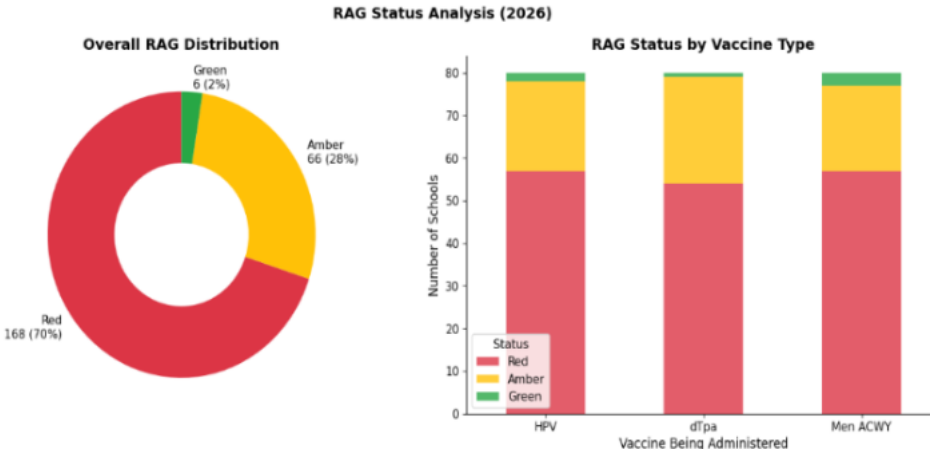


Figure 3. Overall RAG Distribution & RAG Status by Vaccine Type

These graphs show the results of classifying the vaccination rates by school by RAG criteria (red <60%, amber 60-79%, and green >=80%). Of the total 240 school vaccine records, about 168 (70%) were in red status, 66 (28%) were in amber status, and only 6 (2%) were in green. This means that most schools are not reaching the target vaccination rate of 80%. In addition, in the distribution by vaccine type, differences by vaccine type were not noticeable, and schools in red status accounted for the highest proportion. These results show that low vaccination rates are a common problem across programs, not limited to specific schools or specific vaccines. This suggests that more effective participation strategies are needed across programs to increase vaccination participation rates. On the live dashboard, these RAG thresholds flag automatically the moment data is entered, meaning any user can immediately see which schools require urgent attention without having to calculate or categorise manually.

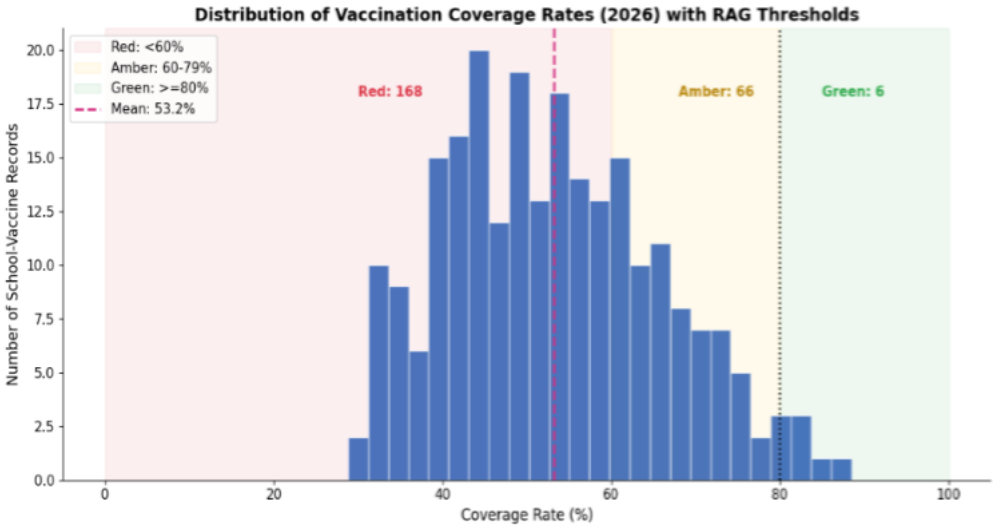


Figure 4. Vaccination Coverage Distribution Across Schools

The histogram above shows the overall distribution of vaccination rates by school. The coverage rate is widely distributed from about 30% to early 80% with an average coverage rate of about 53.2%. In particular, many schools are concentrated in a range between 40% and 60%, and this range mostly falls within the red range (<60%) on the basis of RAG. On the other hand, there are very limited schools in the green range of 80% or more. This distribution shows that not only is the overall average vaccination rate low, but the gap in vaccination rates between schools is also quite large. This suggests that the performance of the vaccination program may vary depending on the school. In addition, it is difficult to assess the program's performance with simple average figures.

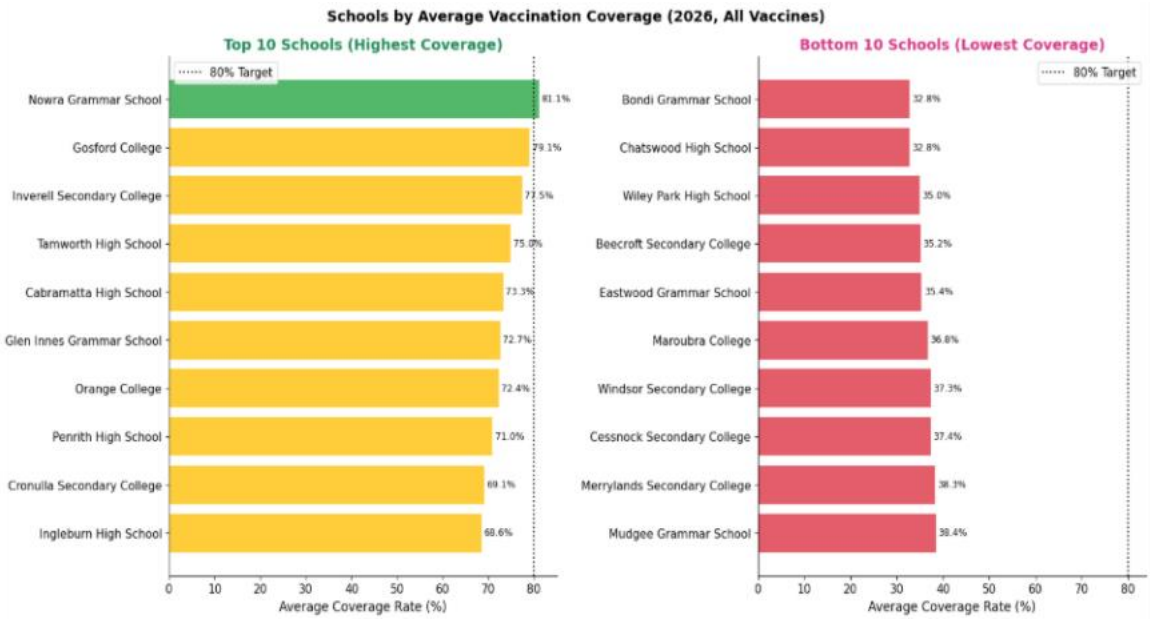


Figure 5. Schools by Average Vaccination Coverage

Figure 5 shows a bar chart comparing between the schools with the highest and lowest vaccination rates. The schools with the highest vaccination rates are Nowra Grammar School (81.1%), the only school that has exceeded the target vaccination rate of 80%. On the other hand, Bondi Grammar School and Chatswood High School have the lowest vaccination rates of about 32.8%. There is a big difference of about 48 percentage points between the top and bottom schools. In addition, most of the top 10 schools have a vaccination rate in the range of 69 to 79%, which is close to the target vaccination rate. These results mean that some schools have been operating relatively successfully, while others have very low participation rates. This suggests the need for a customised approach that considers the situation of each school.

We generated schools with deliberately varied performance levels to test whether the ranking logic would isolate the most critical cases. It did. On the live dashboard, a coordinator can sort by coverage and immediately flag which schools need priority intervention, without any manual analysis.

5.3 Drivers of Vaccination Uptake

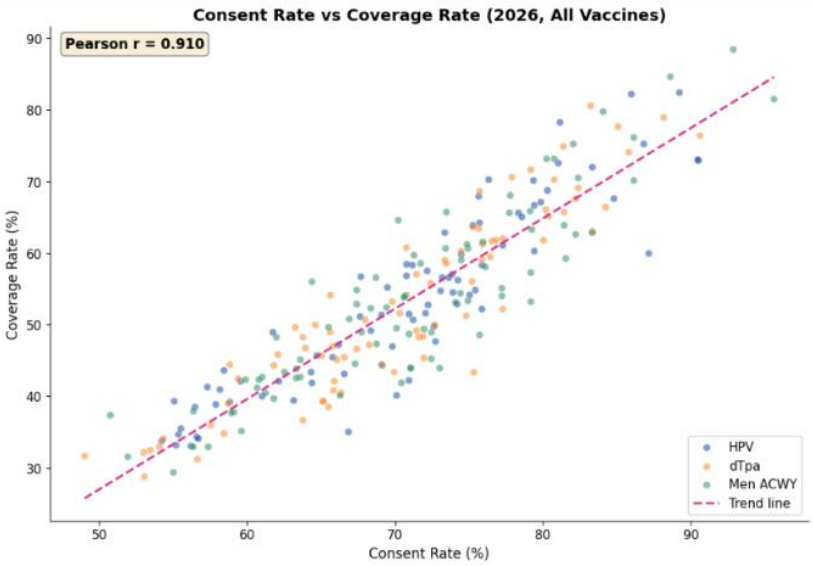


Figure 6. Consent Rate vs Vaccination Rate

This scatterplot shows the relationship between the parental consent rate and the actual vaccination coverage rate. A strong positive relationship appears between the two variables, and the Pearson correlation coefficient is shown as r = 0.91. This indicates that schools with high consent rates tend to have high actual vaccination rates. Conversely, schools with low consent rates tend to have a low vaccination rate. These results show that parents' consent is a key factor in determining the vaccination rate, rather than operating factors such as visits to medical staff or vaccination schedules. This suggests that information provision or communication strategies for parents to increase the consent rate can play an important role in improving the vaccination participation rate.

Having deliberately separate consent and vaccination columns made them distinct variables to test whether the regression would correctly identify consent as the dominant predictor. It did, showing that the dashboard can separate a consent problem from a delivery problem. Because this relationship was built into the synthetic data by design, r = 0.91 demonstrates the pipeline and dashboard rather than a real-world finding. This distinction matters for the dashboard to allow a coordinator can see immediately where to direct resources rather than treating coverage as a single undivided problem.

5.4 Vaccination Process Analysis

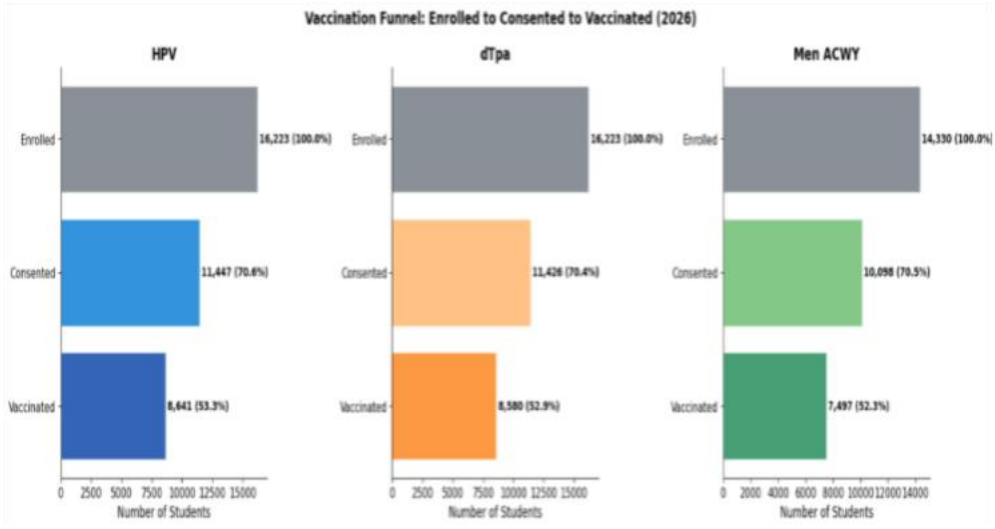


Figure 7. Vaccination Participation Funnel

Figure 7 has 3 separate graphs depicting the stages students leave during vaccination. For the HPV vaccine, about 16,223 students were enrolled as recipients of the vaccination, but only 11,447 (70.6%) of them submitted consent forms, and 8,641 students received the actual vaccination. This means that vaccination participation decreases in two stages. The first decline occurs when the enrolment phase moves to the consent phase, and about 30% of students do not participate at this stage. The second decline occurs at the actual vaccination stage after consent, and some students have not been vaccinated. These results show that the decrease in participation in vaccination is not just a problem of consent rates, but also in the actual operation of the vaccination schedule. We engineered both drop off points deliberately to test whether the model would surface them as two separate issues rather than a single aggregate gap. It did. These are different problems requiring different responses. Consent barriers need parent outreach campaigns, while the post consent gap needs rescheduled follow up visits. The live dashboard makes both visible simultaneously without the user needing to know which question to ask.

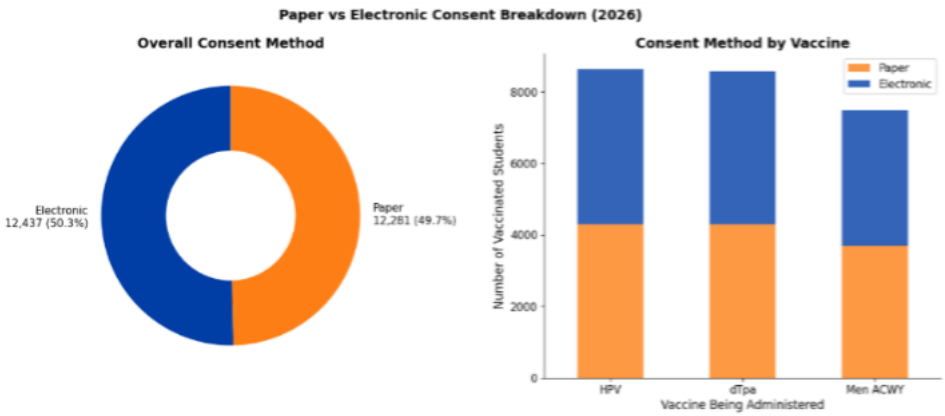


Figure 8. Paper vs Electronic Consent Breakdown

Paper consents account for approximately 49.7% of total vaccinations (Figure 8), with electronic consents making up the remaining 50.3%. We set paper consents above 40% of total vaccinations to validate that the dashboard could quantify the administrative burden of the manual process. On the live dashboard, this metric directly tracks the impact of the MS Forms digital consent rollout. As adoption grows, the paper share should fall, and any coordinator can monitor that shift in real time and flag districts where manual processing remains high.

5.5 Geographic Coverage Analysis

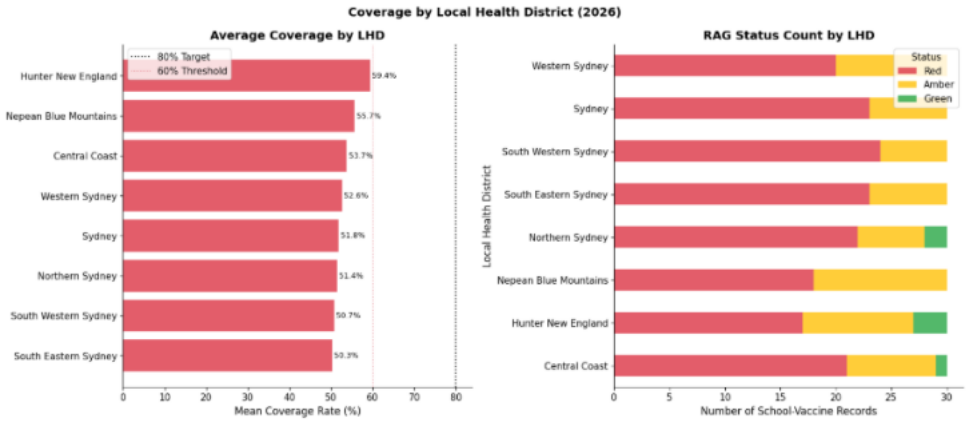


Figure 9. Average Coverage by Local Health District

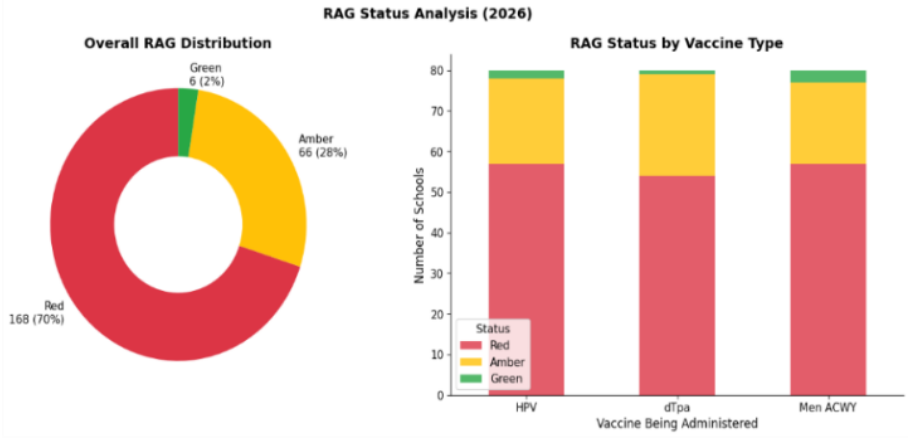


Figure 10. RAG Status Count by LHD

Figures 9 and 10 show the results of comparing the average coverage rates by local health district. It is distributed between about 50.3% and 59.4%. Even though some regions have slightly higher coverage rates than others, all regions still remain below the 80% target vaccination coverage rate. These results show that the problem of the vaccination coverage rate is not limited to a specific region, and it occurs throughout the program. This indicates that a strategy for the overall program, rather than a strategy targeting only a specific region, is necessary to improve the vaccination coverage rate. We assigned schools to 8 real NSW Local Health Districts with deliberately varied performance between districts to test whether the dashboard would surface district specific patterns. It did. We also validated that the Row-Level Security model correctly restricted each coordinator to their own district's data, meaning each user sees only the schools and patterns relevant to their area without needing a centralised report to be generated and distributed.

5.6 Multi-Year Coverage Trends

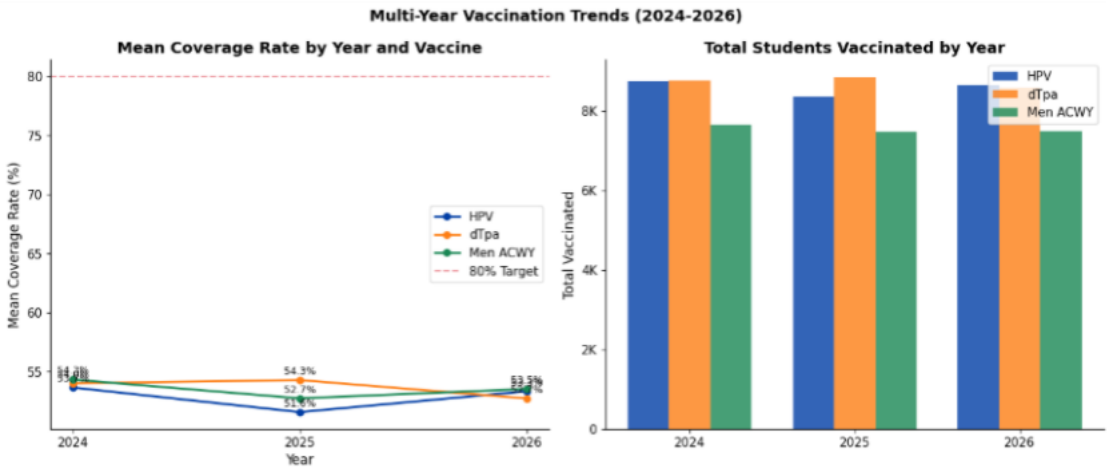


Figure 11. Multi-Year Vaccination Coverage (2024-2026)

Coverage rates show a deliberate year-on-year dip in 2025, informed by our team's research into NCIRS national trends showing consistent post COVID disruption to school based immunization programs, followed by a partial recovery in 2026. The multi-year comparison page detected and displayed this trajectory accurately. Without this view, a coordinator looking at 2026 data alone cannot determine whether 53% represents improvement or continued decline. This is the difference between a static report and a live system, an annual report tells you what happened whereas the dashboard tells you what is happening, allowing any user to flag the program trajectory without waiting for an end-of-year summary.

5.7 Catch up Vaccination Opportunities

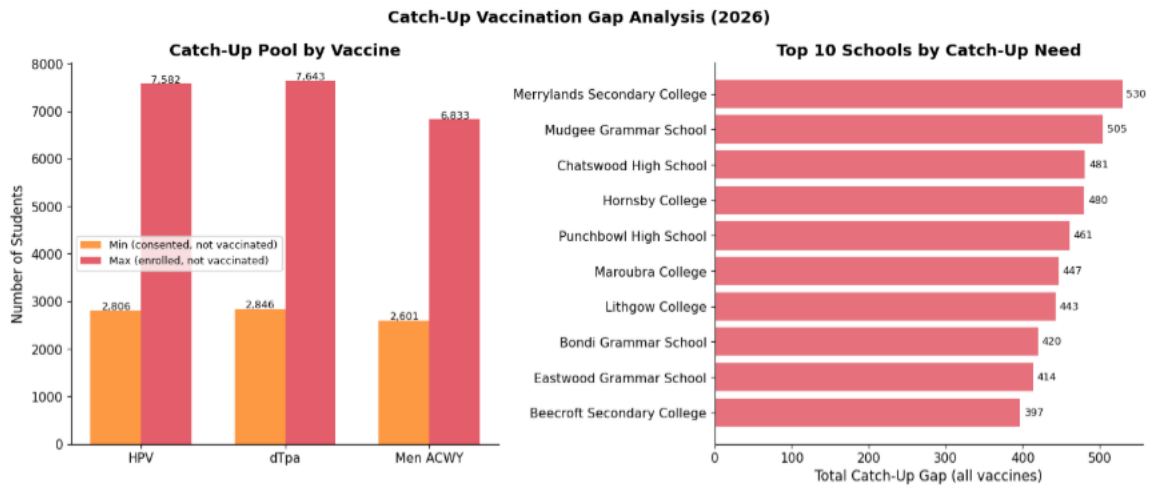


Figure 12. Catch up Vaccination Gap

These graphs show the number of enrolled but not yet vaccinated students. About 7,582 HPV, about 7,643 dTpa, and about 6,833 Men ACWY students have not yet been vaccinated. In particular, there are a significant number of students who have already submitted consent forms but have not received the vaccination. If they provide additional vaccination opportunities, their vaccination coverage rates will improve relatively quickly. We calculated a minimum catch up pool (consented but not vaccinated) and a maximum pool (all enrolled but unvaccinated) to test whether the model would surface these as two distinct actionable targets. It did. The minimum pool represents students reachable in a single follow up visit with no additional outreach required, giving coordinators an immediate action number they can act on without any manual calculation.

According to the bar chart on the right, catch up demand is high at Merrylands Secondary College (530 people), Mudgee Grammar School (505 people), and Chatswood High School (481 people). If additional vaccination programs are operated for these schools first, there is a possibility that the overall coverage rate will be effectively improved.

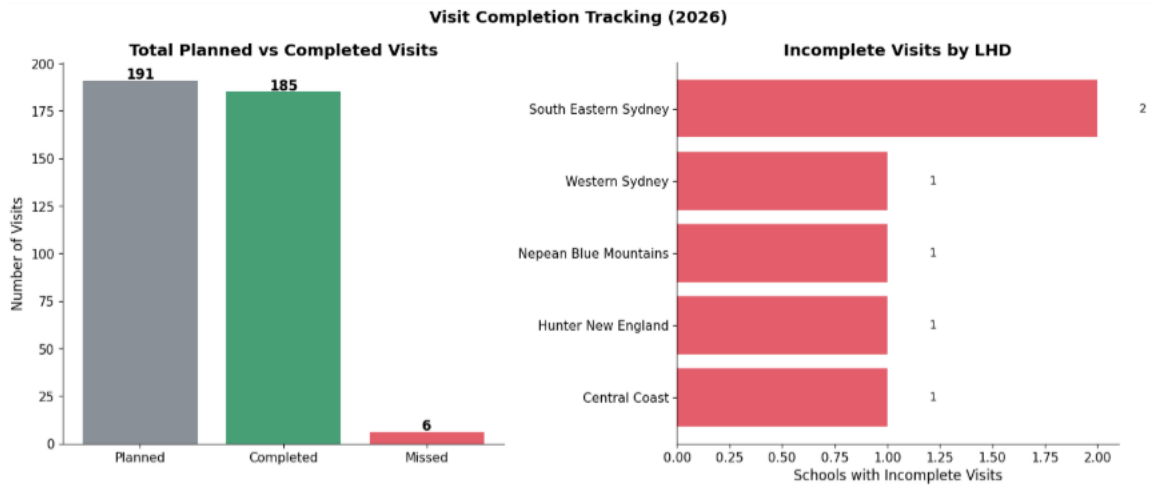


Figure 13. Visit Completion Tracking (Planned vs Completed)

Of 191 planned school visits, 185 were completed, with 6 schools flagged as having incomplete visit schedules across 5 different Local Health Districts. We deliberately set approximately 10% of schools to have planned visits not yet completed to test whether the operational monitoring page would flag them accurately. It did. Vaccination windows close with the school year, a missed second or third visit means those students lose their opportunity entirely. On the live dashboard, any coordinator can open the operational page, see which schools have fallen behind their planned schedule, and arrange rescheduling before the window closes.

5.8 Summary of Key Findings

Diagnostic Question	What We Tested	What the Model Returned	Dashboard Action
How bad is it?	Coverage set below 80% across all schools	53.2% overall, KPI cards surfaced deficit immediately	Coordinator sees gap at a glance on Page 1
Is it everywhere?	Same coverage range across 3 vaccines, varied across 8 LHDs	Systemic across all vaccines, unevenly distributed by district	Filter by vaccine, drill by district, sort schools by coverage
Why is it happening?	Consent and vaccination as separate variables, Two-stage funnel	Consent strongest relationship in synthetic data (r=0.91), two distinct drop-off points surfaced	Direct outreach to consent barriers, reschedule for post consent absentees
Is it getting worse?	Year-on-year decline built into 2025 data	Multi-year page detected dip and partial 2026 recovery	Flag trajectory without waiting for annual report
What can we do now?	Catch up pools and incomplete visit schedules (10% of schools)	Minimum and maximum pools quantified per school, 6 schools flagged	Immediate prioritised action list on operational page
Does the model work?	All characteristics deliberately engineered and tested	Model correctly detected, surfaced, and communicated all issues	Confidence that real data will surface real issues for real coordinators

Across all five dimensions tested, the model consistently detected, surfaced, and communicated the issues we deliberately built into the data, summarised in *Table 1* above. Coverage is critically low at 53.2%, the problem is systemic rather than vaccine specific, consent is the primary bottleneck, geographic disparities exist between districts, and a portion of the gap is recoverable right now through targeted follow up.

Together, these findings answer the core business problem: the state health department currently has no centralised, real time way to see any of this. Coordinators cannot act on what they cannot see, and by the time a spreadsheet is collated or a report is commissioned, vaccination windows may already have closed. The live interactive dashboard addresses this directly. Any coordinator, LHD manager, or school nurse can open it, walk through this exact diagnostic sequence, notice the detail that matters to their area, and flag it for action, without waiting, without requesting, and without needing to interpret the data themselves. That is what the system is built to do, and this analysis confirms it works.

6. Recommendations

Based on the analysis throughout this project, our team has developed a set of practical recommendations for the state health department. These are grouped into short-term (0–3 months), medium-term (3–12 months), and long-term (12+ months) actions, reflecting both urgency and feasibility.

6.1 Short-Term Recommendations (0–3 Months)

6.1.1 Initial Rollout of the Live Power BI Dashboard

The Power BI dashboard developed during this project should be rolled out to state health department coordinators and LHD managers as a priority. At the moment, most coordinators are working off static Excel spreadsheets, which makes real-time monitoring and timely decision-making difficult. In practical terms, the rollout involves several steps. First, the dashboard should be published to a shared state health department workspace through Power BI Service, with Row-Level Security (RLS) configured so each coordinator can only view data relevant to their LHD. Second, within two weeks of deployment, a 60-minute onboarding session should be run for each LHD either by the project team or by a nominated dashboard champion within the LHD. These sessions should focus on navigating the Overview, School Analysis, Operational Monitoring, and Trends pages. Third, user feedback should be collected at the four-week mark and used to refine the dashboard before broader rollout. Once live, coordinators can check vaccination rates, school-level consent figures, and visit completion status in real time without needing to pull together spreadsheets manually.

6.1.2 Prioritising Catch-Up Vaccination Visits

The catch-up analysis identified that 7,582 HPV, 7,643 dTpa, and 6,833 Men ACWY students remain unvaccinated despite being enrolled in the program. A meaningful portion of these students have already submitted consent forms but missed their scheduled vaccination day, meaning they can be reached through a single follow-up visit, with no additional outreach required.

The schools with the highest catch-up need are Merrylands Secondary College (530 students), Mudgee Grammar School (505), Chatswood High School (481), Hornsby College (480), and Punchbowl High School (461). Coordinators should schedule additional visits for these schools within two weeks of this report being reviewed.

The Operational Monitoring page of the dashboard surfaces vaccination gap, incomplete visit status, and completed visit counts side by side, so coordinators can identify which schools need attention and allocate resources without any manual analysis.

School	Total Catch-Up Gap	Priority Action
Merrylands Secondary College	**530 students**	Schedule immediate follow-up visit
Mudgee Grammar School	**505 students**	Schedule immediate follow-up visit
Chatswood High School	**481 students**	Contact school re scheduling
Hornsby College	**480 students**	Contact school re scheduling
Punchbowl High School	**461 students**	Schedule immediate follow-up visit

6.1.3 Resolving Incomplete Visit Schedules Before Term End

The visit tracking analysis found that 6 schools across 5 LHDs have incomplete visit schedules. Because the school immunization program runs within fixed term windows, any visit that does not get rescheduled before term end means those students lose their opportunity entirely.

Coordinators should use the Visit Completion Tracking page to identify affected schools immediately and contact them within one week to lock in rescheduled dates. If a school does not respond within two weeks, the issue should be escalated to the LHD manager. Visit completion status should be updated in the database as soon as each visit is done, so the dashboard always reflects current progress.

6.1.4 Transitioning to Standardized Digital Data Submission

The current reporting system relies heavily on inconsistent Excel-based submissions, which creates data quality problems and processing delays. The Microsoft Forms submission system built during this project should become the standard method for all school immunization reporting across NSW.

The transition process should include distributing a one-to-two page instruction guide to school health teams explaining how to complete and submit the form. Since submissions flow automatically through Power Automate into the Azure SQL staging table, manual data entry at the health team level becomes unnecessary.

The target should be full digital adoption across all schools by the end of the current term, tracked LHD by LHD. Schools that have not transitioned by that point should receive direct support. Once digital adoption is confirmed, the existing email and Excel reporting process can be phased out.

6.2 Medium-Term Recommendations (3–12 Months)

6.2.1 Developing a Consent-Focused Parent Communication Strategy

In the synthetic prototype data, the analysis showed a Pearson correlation of r = 0.91 between consent rate and vaccination coverage; if this pattern holds in real data, parental consent would be the single highest-leverage point for improving program outcomes. The dashboard visualizes consent rate alongside vaccination rate at the school level, allowing coordinators to distinguish between schools where coverage is low because of an operational issue versus schools where the problem starts at the consent stage.

Practically, schools with a consent rate below 65% should be prioritized for targeted outreach. Translated materials should be prepared for schools with significant non-English-speaking communities. A reminder should be sent to parents who have not submitted consent forms within 14 days of the initial distribution, using whatever communication channel the school already has in place, whether that is a school app, SMS, or email.

Presenting the program at P&C meetings or school events can also help raise awareness and address parental concerns directly. This kind of targeted approach is likely to deliver better results than simply increasing the number of scheduled visits.

6.2.2 Applying Vaccine-Specific Coverage Thresholds and Alerts

At the moment, the dashboard applies a single 80% benchmark across all three vaccines. Given that herd immunity thresholds vary. HPV requires 80–85%, dTpa requires 92–95%, and Men ACWY requires 85–90%. This single threshold understates the gap for dTpa and Men ACWY in particular. The figures used in this report show a 28percentage point gap for HPV, 12.2 points for Men ACWY, and 6.5 points for dTpa, so the situation is more serious than a uniform benchmark suggests.

The recommendation is to update the dashboard to apply vaccine-specific Red thresholds and configure Power BI alerts so that when a school's coverage for any vaccine drops below its respective threshold, an automatic email notification is sent to the relevant LHD coordinator. This removes the need for coordinators to check the dashboard manually and enables a more proactive response.

Vaccine	Disease Target	Herd Immunity Threshold	Coverage used in this report
HPV	HPV-related cancers	80–85%	~52% (2024, synthetic)
dTpa	Diphtheria, Tetanus, Pertussis	92–95%	85.5% (2023)
Men ACWY	Meningococcal disease	85–90%	72.8% (2023)

6.2.3 Expanding Electronic Consent

Approximately 49.7% of vaccinations in the dataset were linked to paper-based consent, which adds significant administrative work for health teams and slows down data processing. A concrete target of reducing paper consent to below 20% of total vaccinations within 12 months should be set.

Steps toward this include actively promoting the electronic consent option through school communication channels and exploring whether the system can be linked to platforms schools already use, such as Compass or Sentral. The paper vs. electronic consent breakdown on the dashboard should be reviewed monthly by LHD managers, and electronic consent rate should be formally included as an operational KPI. Any LHD where paper consent remains above 40% at the six-month mark should be flagged for additional support.

6.3 Long-Term Recommendations (12+ Months)

6.3.1 Introducing Predictive Analytics and Machine Learning

Once real operational data has accumulated over two or more years, there is a genuine opportunity to move beyond descriptive analytics into predictive modelling. Three approaches are worth pursuing. A linear regression model could use early-term consent rates, historical coverage, and school demographics to forecast end-of-year vaccination rates before the immunization season is over. A classification model, such as Random Forest or Gradient Boosting, could flag schools at high risk of falling into Red status at the start of each term, enabling proactive resource allocation rather than reactive catch-up. A time-series model such as ARIMA or Prophet could project multi-year vaccination trends at the LHD and state level. These models could be integrated directly into the Power BI dashboard, displaying predicted coverage alongside actual figures and automatically highlighting schools where a Red outcome is expected. A reasonable success threshold for any model before full deployment would be 80% prediction accuracy, validated against a held-out portion of the real dataset.

6.3.2 Scaling the System Across NSW

The current prototype covers 80 schools across 8 LHDs. If the pilot period, defined as the first six months of live operation, meets the criteria of 70% or above dashboard usage, 95% or above data accuracy, and a user satisfaction score of at least 4 out of 5, the system should proceed to a full NSW rollout.

For production deployment, the Azure SQL instance should be upgraded from the current free-tier configuration to a General Purpose or higher tier, with automated backups, private endpoint access, and Azure Defender for SQL enabled. The RLS framework already in place can accommodate additional LHDs without structural changes, and the centralised database and Power BI Service environment are capable of handling a much larger volume of schools and users. Over time, the state health department should look to formally designate a data stewardship role internally to take ownership of dashboard maintenance, data quality monitoring, and access governance.

6.4. Summary

The following table provides a consolidated overview of all recommendations by priority and timeline.

Recommendation	Priority	Timeline	Responsible Party
Deploy Power BI dashboard to coordinators	High	0–1 month	IT / LHD Managers
Prioritise catch-up visits for low-coverage schools	High	0–1 month	Field Coordinators
Resolve incomplete visit schedules	High	0–2 months	Field Coordinators
Introduce standardised MS Forms submission	Medium	0–3 months	Program Management Team
Consent-focused parent outreach strategy	Medium	3–6 months	LHD Coordinators
Implement vaccine-specific thresholds & alerts	Medium	3–6 months	Data / IT Team
Expand electronic consent adoption	Medium	6–12 months	Program Management Team
Enhance long-term vaccination trend analysis	Long-term	12+ months	Analytics Team
Gradual expansion across additional NSW LHDs	Long-term	Long-term	State Health Department Management

7. Deployment

7.1 Prototype Deployment Status

By the end of the project, the group had achieved a functioning prototype deployment that demonstrated the main reporting workflow required by the state health department. The system was able to capture submission data through Microsoft Forms, write responses into Excel on OneDrive, detect new rows using Power Automate, transfer the submitted data into Azure SQL, and prepare that data for reporting in Power BI. This meant that the project moved beyond a static dashboard concept and into a working end-to-end prototype.

At the same time, the deployed system should be understood as a prototype rather than a fully production-ready solution. The core workflow was successfully demonstrated, but final deployment still depends on stronger service-side configuration, more stable cloud access, and the replacement of dummy inputs with real operational data. In this sense, the project confirmed feasibility and practical value, while also making clear what still needs to be completed before operational rollout.

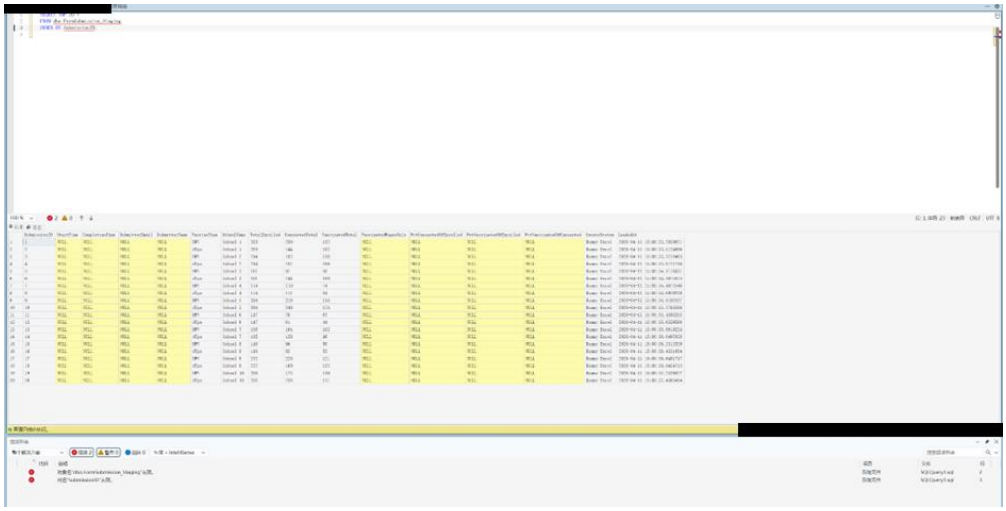


Figure 7.1: Successful insertion of submitted rows into the Azure SQL staging table during prototype deployment.

7.2 Final Deployment Architecture

The final deployment architecture used in this project followed the workflow below:

Microsoft Forms → Excel on OneDrive → Power Automate → Azure SQL → Power BI

This architecture supported the project's main objective of improving how school immunization data is collected, structured, and visualised. Microsoft Forms provided a standardised submission layer for staff. Excel on OneDrive captured the responses automatically. Power Automate acted as the automation layer by detecting new rows and transferring them into the backend. Azure SQL provided structured relational storage and validation support. Power BI then used the processed data for reporting, modelling, and dashboarding.

A key deployment decision was the use of a staging table. Instead of writing raw form responses directly into the final reporting tables, each submission was inserted into StagingTable first. This preserved the original row, reduced the risk of data loss during automation, and created a safe intermediate layer for validation and promotion into the final production tables. This design was especially important because incoming data could contain formatting or type issues that needed to be handled before the data was ready for reporting.

Stage	Tool	Role in the System
1	Microsoft Forms	Captures staff vaccination submissions
2	Excel on OneDrive	Stores form responses automatically
3	Power Automate	Detects new rows and transfers data into Azure SQL
4	Azure SQL	Stores staged data for structured modelling and validation
5	Power BI	Uses SQL data for reporting, modelling, and dashboarding

Figure 7.2: Final prototype deployment architecture from Microsoft Forms through to Power BI.

Power Automate Field Mapping			
Microsoft Forms / Excel -> Azure SQL staging table			
Excel / Form Column	**Suggested SQL Field**	**Type**	**Action / Rule**
Id	SubmissionID	INT	Use as primary key
Start time	StartTime	DATETIME2	Convert to datetime
Completion time	CompletionTime	DATETIME2	Convert to datetime
Email	SubmitterEmail	NVARCHAR(255)	Allow null if blank
Name	SubmitterName	NVARCHAR(255)	Allow null if blank
Vaccine being administered	VaccineType	NVARCHAR(50)	Standardise to HPV / dTpa / Men ACWY
School name	SchoolName	NVARCHAR(255)	Keep spelling consistent
Total enrolled	TotalEnrolled	INT	Must be numeric
Consented (paper and electronic)	ConsentedTotal	INT	Must be numeric
Total vaccinated (including paper constents)	VaccinatedTotal	INT	Must be numeric
Number vaccinated (paper consents only)	VaccinatedPaperOnly	INT	Nullable
% consented (of enrolled)	PctConsentedOfEnrolled	DECIMAL(5,2)	Can be calculated later
% vaccinated (of enrolled)	PctVaccinatedOfEnrolled	DECIMAL(5,2)	Can be calculated later
% vaccinated (of consented)	PctVaccinatedOfConsented	DECIMAL(5,2)	Can be calculated later

Figure 7.3: Field mapping used to align form and Excel responses with the Azure SQL staging structure.

7.3 Deployment Issues

Although the prototype workflow was successfully implemented, deployment revealed several practical issues. The first was the need to revise the planned system architecture. Earlier project planning assumed a backend approach centred on Microsoft Access, but this was later replaced with Azure SQL and Power Automate because the original setup proved impractical within the available environment. While this change improved the final structure of the solution, it also required redesign work and additional setup time.

A second issue involved database setup and infrastructure configuration. Before the database could be used reliably, firewall settings, user permissions, administrative access, and connectivity had to be configured correctly. These tasks were more time-consuming than expected and showed that deployment depends not only on schema design, but also on stable access and environment management.

A third issue concerned data integration. The form fields, Excel response structure, Power Automate mappings, and SQL table schema all had to align exactly. Small inconsistencies such as mismatched columns, invalid date formats, and type conversion problems were enough to interrupt the pipeline. This made the staging-table design especially valuable, because it allowed raw data to be preserved safely before further processing.

A final issue emerged at the live reporting stage. The dashboard model could be viewed successfully in Power BI Desktop, which confirmed that the reporting logic was working. However, final service-side deployment still required stable data source credentials, cloud availability, and production-ready service configuration. This showed that a locally functioning prototype and a fully live operational deployment are not the same thing, and that both stages need to be considered separately.



Figure 7.4: Azure SQL networking configuration used to support team access and service connectivity.

7.4 Where to Go from Here

The next major step is to replace the prototype or dummy inputs with live operational data from the state health department. At present, the system successfully demonstrates the workflow and reporting concept, but a production deployment would need to ingest genuine staff submissions and operate against real immunization records on an ongoing basis. This would allow the dashboard to support actual operational monitoring rather than prototype testing.

The second priority is to strengthen validation, governance, and service configuration. This includes finalising Power BI Service credentials, improving live refresh reliability, strengthening form-level and database-level validation rules, and defining clearer ownership for access management and maintenance. These steps are important if the system is to function as an ongoing operational tool rather than a project prototype.

A further area for future work is system scale and organisational rollout. The current prototype covers 80 schools across 8 Local Health Districts and already includes row-level security and a centralised relational model. This means the design can be extended further once the live deployment environment is stabilised. Over time, the state health department could expand the system into a broader operational reporting platform, supported by regular maintenance, stronger monitoring, and formal internal ownership.

7.5 Reflection and Lessons Learned

This project showed that a successful analytics solution depends on far more than dashboard design alone. The group learned that backend structure, validation, permissions, and deployment planning are just as important as front-end reporting. A dashboard can only support decision-making effectively when the submission process, storage layer, transformation logic, and access rules all work together consistently.

The project also showed the value of flexibility during implementation. When the original backend approach proved impractical, the team revised the architecture and adopted a more workable prototype design using Azure SQL and Power Automate. Although this created short-term disruption, it resulted in a stronger and more scalable workflow that aligned better with the client's long-term needs.

Finally, the group learned that deployment should be treated as a continuing process rather than a single final step. Even after the core pipeline and dashboards are functioning, production readiness still depends on stable infrastructure, access governance, service configuration, and continued refinement. The final prototype therefore serves both as a working proof of concept and as a practical foundation for future implementation.

8. Difficulties Encountered

8.1 Challenges Encountered During Early Development Phases

Lack of Communication with Supervisor:

We attempted to contact the supervisor multiple times via email to clarify both business and technical requirements. However, we did not receive any response. This created uncertainty regarding the expectations of the project and limited our ability to validate our approach.

Use of Microsoft Access:
Initially, the project required the use of Microsoft Access as the database system. However, it was not included in the university subscription plan and required an additional paid license. This created a limitation for our team, as we were unable to access or use the required software directly. As a result, we had to spend extra time researching and evaluating alternative database solutions that could meet the project requirements.

Database Setup and Configuration Issues:
Setting up the database environment proved to be challenging due to several configuration requirements. These included setting up firewall rules, assigning administrator access, resolving login issues, and establishing a connection between Python and the database. Each of these steps required troubleshooting and adjustments before the system functioned correctly.

Server Connectivity Issues:
During the setup process, the database server could not be detected at one point, which delayed progress. To resolve this, we refreshed the system and used diagnostic tools such as *nslookup* and *ping* to verify whether the server was active and reachable. This process required multiple attempts before a successful connection was established.

Data Import and Integration Challenges:
Importing the dataset into the staging table was one of the most time-consuming challenges, taking approximately 4 hours of troubleshooting. The column names in the CSV file needed to exactly match the database schema when executing INSERT queries. Initial attempts resulted in rows being created with NULL values due to mismatched columns. Additional issues included file access errors, missing file paths and system crashes while processes were running. Significant effort was required to debug these issues, ensure data consistency, and validate the dataset before moving clean data into the final table.

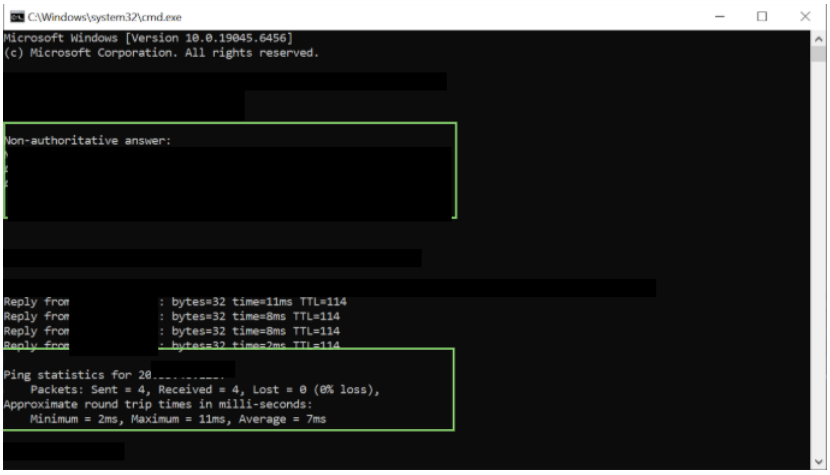
8.2 Mitigation Strategies During Early Development Phases

Switch to Azure SQL:
To overcome the limitation of Microsoft Access, we adopted Azure SQL Database, which is a cloud-based, reliable, and user-friendly alternative. It operates under a Platform as a Service (PaaS) model, meaning the cloud provider manages infrastructure, maintenance, and updates, allowing users to focus only on database usage and development. We are currently using the free trial version.

Proceeding with Available Requirements:
Due to the lack of supervisor communication, we relied on the project summary document provided on Canvas to guide our understanding of the requirements.

Resolving Configuration Issues:
We carefully configured firewall rules, ensured correct administrator permissions and troubleshot login and connectivity issues to successfully set up the database environment.

Troubleshooting Server Issues:
Server connectivity problems were resolved by restarting the system and using network diagnostic commands to confirm that the server was active and reachable.



Screenshot 10: Command prompt output showing successful DNS lookup and network connectivity to the Azure SQL database server (`your-server.database.windows.net`), with 0% packet loss.

Fixing Data Import Errors:
Loading the dataset into the staging table required approximately four hours of troubleshooting. We ensured that column names in the CSV file exactly matched the database schema when executing INSERT queries. Earlier attempts resulted in NULL values due to mismatched columns. Additional issues, such as file access errors and system crashes, were resolved through repeated testing and corrections. Data quality checks were also performed to identify missing values and inconsistencies before transferring clean data into the final table.

8.3 Challenges Encountered During Final Development Phases

Azure SQL to Power BI Connection Issues:
During the final development stage, our team experienced difficulties while connecting the Azure SQL database to Power BI. Though at that stage the database had already been created and filled, Power BI required the correct server details, user permissions, database roles, and Azure firewall settings before the dashboard could properly access the data for the live pages to be developed. This delayed our start to building the dashboard thus also delaying our testing phase.

Collaboration/Version Control Issues:
Collaboration in regard to working with Power BI and Azure became difficult as updates did not always appear consistently for all team members. Some members were able to see the most updated version while others could not, which forced us to adjust our approach when developing the dashboard. In order to avoid overwriting work, our team had to ensure only one person edited the dashboard at a time.

Power BI Trial/Subscription Limitations:
One of the major issues our team faced was related to software access constraints with Power BI. The free trial period shortened our allocated time for developing the dashboard which inevitably ended up forcing our team to pay a monthly subscription in order to continue accessing and finalizing the work.

Shortened Testing/Refinement Timeframe:
As a result of the time spent resolving the previously mentioned issues, less time was available for dashboard testing and refinement. Our team had to prioritize core dashboard functionality to ensure that the most critical features work as intended prior to final submission.

8.4 Mitigation Strategies During Final Development Phases

Resolving Azure SQL to Power BI Connection Issues:
To resolve the connection issues, our team reviewed the Azure SQL server details, checked and resolved the user permissions, and adjusted the setting to enable the database access for Power BI. Once these changes were made, we quickly verified the fix through a simple trial graph to ensure the data was being called correctly.

Managing Collaboration and Version Control:
The main issue with version conflicts was addressed by coordinating the dashboard development amongst the team and limiting edits to a single person at a time. This helped to ensure progress would not be overwritten.

Managing Power BI Access Limitations:
Though initially, our team adjusted the deadline to complete the dashboard pages before the free trial expired, we were unaware that viewing the dashboard in any manner would be restricted. This forced us to pay a monthly subscription to be able to access our work for the presentation and final report as we still needed the ability to reference our live pages in order to complete all our requirements.

Prioritizing Essential Testing and Refinements:
As a result of all the time constraining issues, our team focused on ensuring that the most important dashboard functions were working properly before proceeding to fine tuning and testing the dashboard. Though this limited the depth to which we were able to scrutinize our product, it ensured that the functionality of the system was operational, reliable, and ready to be presented.

9. Conclusion

The project aims to improve the operational process of the state health department. The current systems rely on multiple Excel files, making it difficult to manage vaccination status in real time. To improve this, the team built a relationship database, a Microsoft Forms, a real-time Power Automation data pipeline, and a Power BI dashboard. Through this, it was possible to incorporate the report process that was previously dispersed into one centralized system.

Based on data from 80 schools and 8 local health distribution areas, the overall vaccination rate was 53%, below the target of 80%. These results showed that the problem occurs across common areas in various regions, not in particular schools or areas. In addition, the strong correlation between parental consent and vaccination rates in the synthetic data illustrates why consent is worth monitoring. In addition, it appears that the recommended vaccination criteria have not been reached, and there is a need for a more integrated operating strategy.

The system developed through the project addressed the limitations of the existing Excel-based method in operational management. Through the Row-Lastel Security features, users can only check their local data, and validation check procedures are reflected in the dashboard. These supporting functions can be used as a real operational monitoring tool beyond simple reporting tools.

Although the current prototype was built using dummy data, the entire system architecture was designed to integrate with future operational data. If the appropriate distribution and additional development are implemented, this system is expected to support vaccination status and enable the state health department to be more efficiently monitored and respond faster to schools.

10. Appendix

Appendix A - Overview dashboard





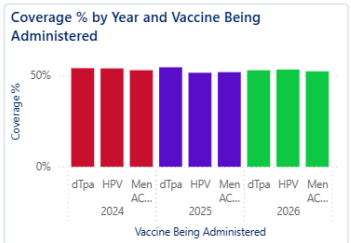


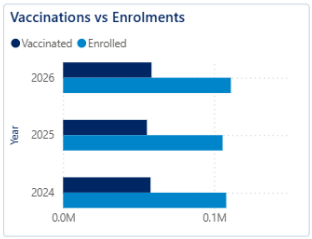
Appendix B – School level analysis dashboard

School Level Analytics

School Name: Enter School Name

2024 | 2025 | 2026





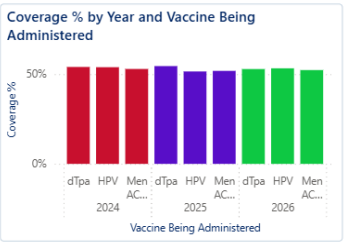
53.7% Coverage %	71.8% Consent Rate %
74.8% Vaccinated of Consented %	552 Visits Completed

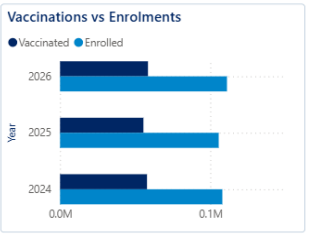
School Name	Enrolled	Vaccinated	Coverage %	Consent Rate %	Visi
Armidale Grammar School	886	582	65.7%	80.8%	
Ashfield High School	1206	614	50.9%	73.5%	
Auburn Secondary College	2418	1794	74.2%	83.3%	
Bankstown High School	1286	604	47.0%	70.5%	
Bathurst Secondary College	2031	1086	53.5%	73.0%	
Total	**108007**	**57865**	**53.7%**	**71.8%**	

School Level Analytics

School Name: Enter School Name

2024 | 2025 | 2026





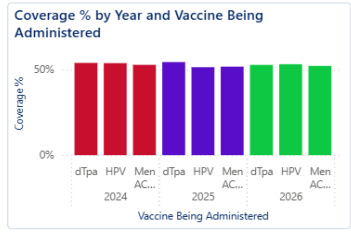
52.7% Coverage %	70.4% Consent Rate %
74.8% Vaccinated of Consented %	543 Visits Completed

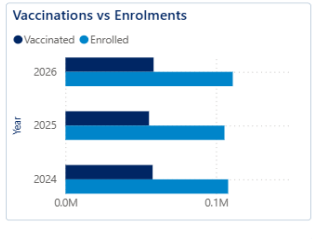
School Name	Enrolled	Vaccinated	Coverage %	Consent Rate %	Visi
Armidale Grammar School	864	498	57.6%	75.0%	
Ashfield High School	1194	564	47.2%	63.0%	
Auburn Secondary College	2319	1485	64.0%	77.9%	
Bankstown High School	1302	622	47.8%	68.5%	
Bathurst Secondary College	2076	1095	52.7%	68.5%	
Total	**105570**	**55473**	**52.7%**	**70.4%**	

School Level Analytics

School Name: Enter School Name

2024 | 2025 | 2026

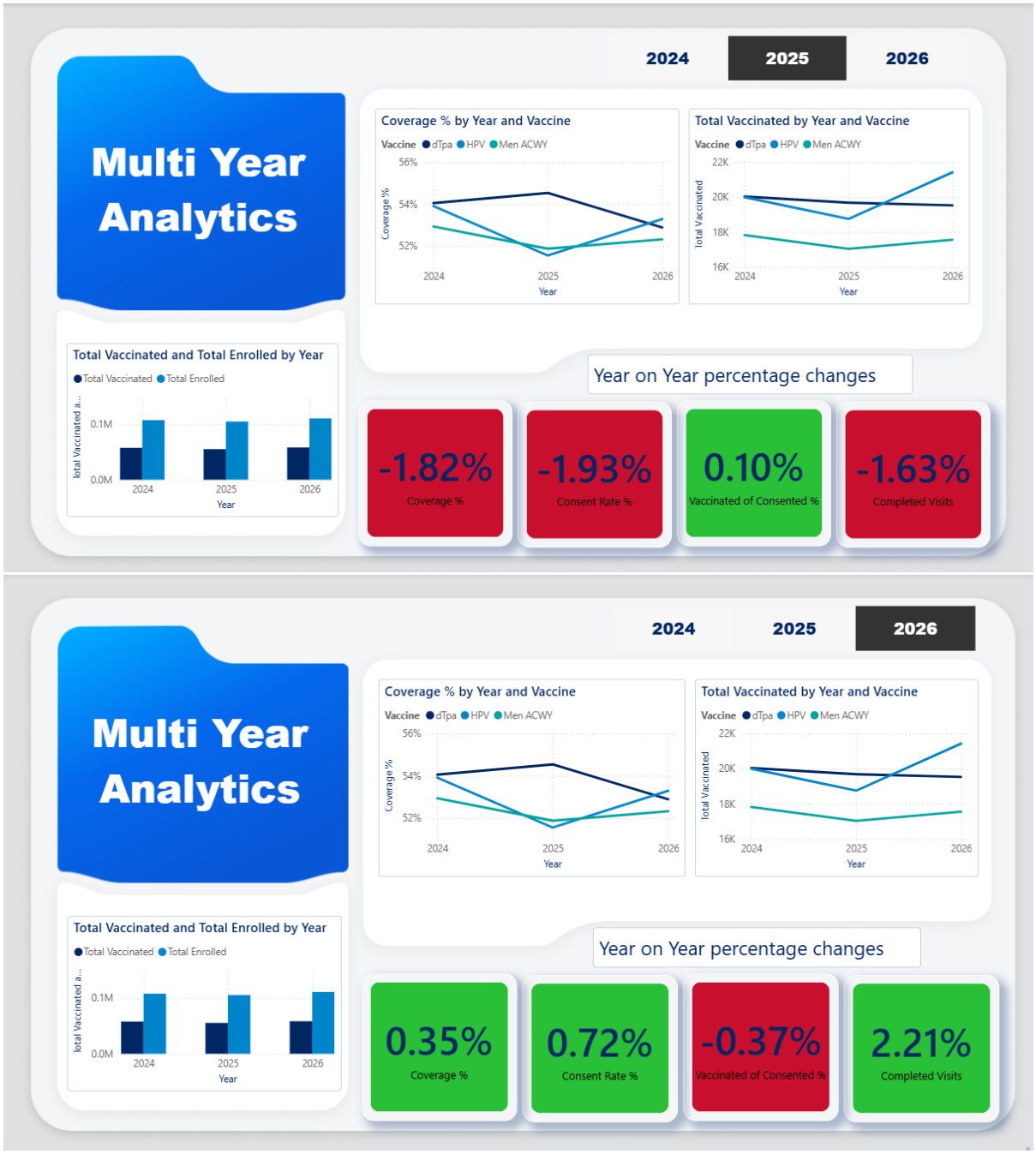




52.9% Coverage %	70.9% Consent Rate %
74.6% Vaccinated of Consented %	555 Visits Completed

School Name	Enrolled	Vaccinated	Coverage %	Consent Rate %	Visi
A	2280	1140			
Armidale Grammar School	830	490	59.0%	75.4%	
Ashfield High School	1166	554	47.5%	66.9%	
Auburn Secondary College	2280	1506	66.1%	76.7%	
Bankstown High School	1244	502	40.4%	67.8%	
Total	**111077**	**58508**	**52.9%**	**70.9%**	

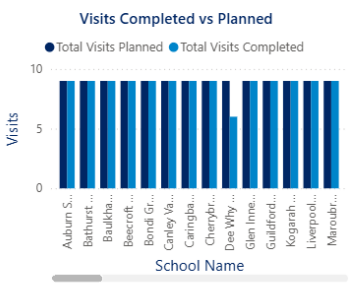
Appendix C – Multiple year analysis dashboard

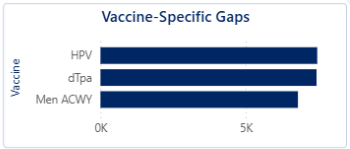


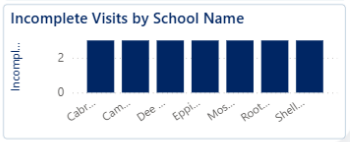
Appendix D – Multiple year analysis dashboard

Operational Monitoring

2024 | 2025 | 2026







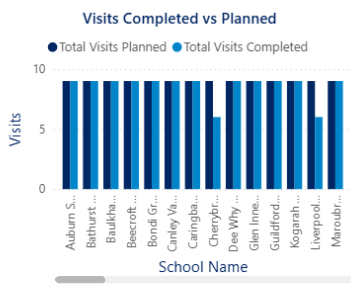
Schools Below Threshold

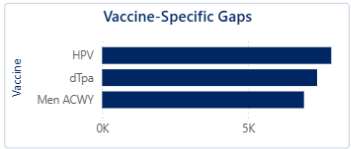
School Name	Coverage %	Status
Windsor Secondary College	33.8%	Red
Bondi Grammar School	34.5%	Red
Cessnock Secondary	34.6%	Red

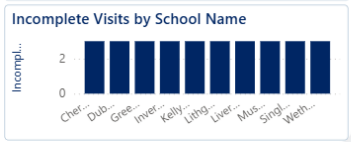
573	552	96.3%	80
Total Visits Planned	Total Visits Completed	Visit Completion Rate	Schools Below Threshold

Operational Monitoring

2024 | 2025 | 2026







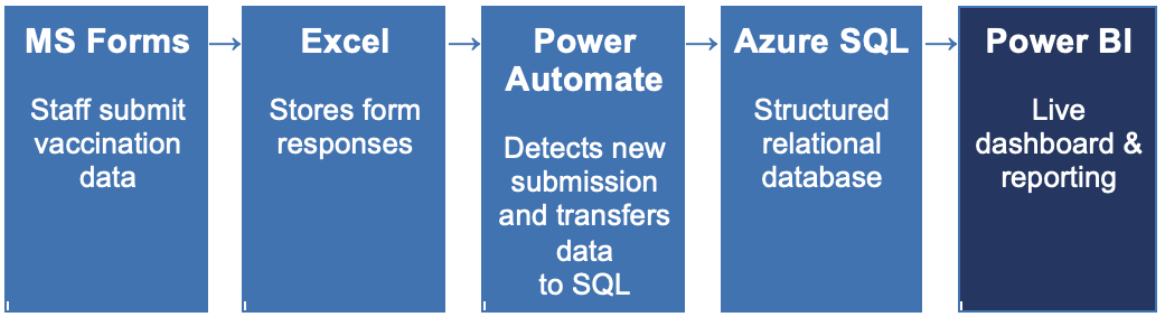
Schools Below Threshold

School Name	Coverage %	Status
Wiley Park High School	34.3%	Red
Beecroft Secondary College	35.2%	Red
Windsor Secondary	35.5%	Red

573	543	94.8%	80
Total Visits Planned	Total Visits Completed	Visit Completion Rate	Schools Below Threshold



Appendix E – Data pipeline overview



Appendix F – Data dictionary

Attribute	Description
RecordId	Unique identifier for each vaccination record
Submission Time	Date and time the form was submitted
Email	Email address linked to the submission
Year	Reporting year
Year Group	Student year group targeted for vaccination
Vaccine Being Administered	Vaccine type administered during the visit
School Name	Name of the school
Suburb	School suburb location
Postcode	School postcode
Local Health District	Local Health District (LHD) linked to the school
Is SSP School	Indicates whether the school is an SSP school
Total Enrolled	Total number of enrolled students
Consented (Paper and Electronic)	Total number of students with parental consent
Total Vaccinated (Including Paper Consents)	Total number of vaccinated students
Number Vaccinated (Paper Consents Only)	Number vaccinated using paper consent forms
Planned Visits	Total planned vaccination visits
Visits Completed	Total completed vaccination visits
Visit 1 Date	Date of first vaccination visit
Visit 2 Date	Date of second vaccination visit
Visit 3 Date	Date of third vaccination visit
Catch-Up Min Expected	Minimum expected catch-up vaccinations
Catch-Up Max Expected	Maximum expected catch-up vaccinations
% Consented (of Enrolled)	Percentage of consented students
% Vaccinated (of Enrolled)	Percentage vaccinated out of enrolled students
% Vaccinated (of Consented)	Percentage vaccinated out of consented students
Number Vaccinated (Electronic Consents Only)	Vaccinated students using electronic consent
Coverage Status (RAG)	Coverage classification using RAG system
Visit Status	Vaccination visits completion status

Appendix H Microsoft form

Vaccination information form

To be filled out by nurse for vaccines administered at school

When you submit this form, it will not automatically collect your details like name and email address unless you provide it yourself.

* Required

1. School Name *

Enter your answer

2. Vaccine Type *

- ◯ HPV
- ◯ MenACWY
- ◯ dTpa

3. Visit Date *

Please input date (M/d/yyyy)

4. Visit Number *

The value must be a number

5. Year group *

- ◯ 7
- ◯ 10

6. Total Enrolled *

Please enter a number greater than 0

7. Total Consented *

The value must be a number

8. Total Vaccinated *

The value must be a number

9. Paper Consents *

The value must be a number

Submit

Never give out your password. Report abuse